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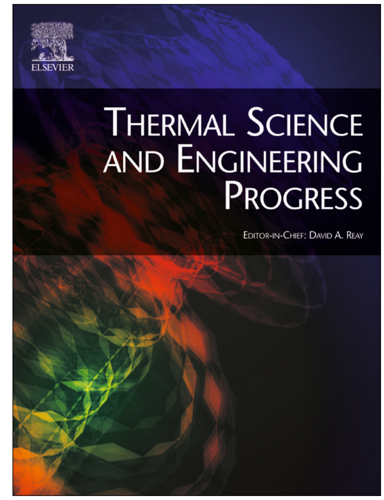

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Enhancing Outdoor Comfort through Tensile Membrane Structures and Pavement Surfaces: A Case Study Report in Évora, Portugal

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Abstract

The Universal Thermal Climate Index (UTCI), developed in 2009 through collaboration among experts in human thermophysiology, aims to evaluate the relationship between the outdoor environment and human well-being. The case study focused on assessing the impact of shadowing and thermal masses on occupant outdoor thermal comfort as measured by the UTCI. The study specifically examined the effects of various designs of tensile membrane structures (TMS) on a U-shaped patio enclosed by nine maritime containers arranged in three groups of three levels in Évora, Portugal. The model tested fourteen TMS with varying slopes, overlap angles, lengths, repetitions, and four different pavements with distinct thermal mass layouts. The methodology employed dynamic simulation software, the energy engine EnergyPlus 2022, to assess the monthly variation of UTCI temperatures, comparing them to the scenario of an uncovered patio for each pavement configuration. The best-performing patio follows a TMS with a slope of 38.2º, an overlap of 74.4º, a length of 3.88 m, and five repetitions, accompanied by a full-concrete pavement. During winter (>9ºC), the best-performing TMS underperforms against the uncovered patio by adding 1.1ºC but during summer (>26ºC), outperforms by reducing user discomfort by 9.4ºC. The global temperature deviation with the best-performing TMS sets at 3.3ºC, where the uncovered patio at 11.8ºC, is out of the comfort gap (9ºC to 26ºC) by 8.3ºC, benefiting user comfort across the year. The study highlights the potential of optimizing outdoor spaces based on UTCI measurements to create comfortable and user-friendly environments.

Keywords: Universal Thermal Climate Index, dynamic energy simulation, outdoor user comfort, tensile membrane structures, design optimization, shadow performance

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Acronyms and Abbreviation:

UTCI - Universal Thermal Climate Index

UTCI ΔT - Universal Thermal Climate Index change in temperature

ΔT - Change in temperature

TMS – Tensile Membrane Structures

PS – Pavement surfaces

PET – Physiologically Equivalent Temperature

PTFE – Polytetrafluoroethylene

BES – Building Energy Simulation

LB – Ladybug-Tools

EM – ENVI-met software[2]

MET – Metabolic rate

MRT – Mean Radiant Temperature

OTC – Outdoor Thermal Comfort

TCP – Thermal Comfort Percent

HSP – Heat Sensation Percent

CSP – Cold Sensation Percent

1. Introduction

The world is experiencing a persistent challenge of human migration and housing shortages [1], particularly in economically and technologically advanced regions. In the best-case scenarios, the constant influx of tourists, retirees, and wealthy visa holders has exacerbated the issue [2]. These pressures force cities to accommodate both young and aging populations, particularly in Europe. In worse situations, people are forced to migrate due to the effects of climate change, territorial disputes, social unrest, and individual persecution.

EU member states have taken the lead in providing shelter and housing for those lacking opportunities or simple teleworking accommodations. As a response, the EU has implemented policies promoting social stability, economic strength, and integration opportunities, including creating new housing areas with carefully designed public urban spaces [3].

[2] ENVI-met is a high-resolution 3D modeling software that accurately simulates complex microclimatic processes.

Based on the considerations above, the research team identified a need for lightweight architectural solutions. These solutions should feature defined and enclosed outdoor spaces that facilitate social interaction while adhering to standard outdoor comfort requirements. The case study, which resorts to a tensile membrane structure (TMS) outdoor covering, focuses on the region of Évora, Portugal, due to its similarities with other locations along the Mediterranean landscape.

Évora is situated in Alto Alentejo (38°34′00″ N, 7°54'48″ W), a southern region of the country that lies between Estremadura (mainly the Lisbon area) and Ribatejo (above the Tagus River) to the north and Baixo Alentejo to the south, bordering Algarve. The climate of the region can be described as warm temperate with dry, hot summers and a Mediterranean climate (Csa) according to the Köppen-Geiger classification system (1980-2016) [4], [5]. The winter solstice angle reaches 27.6º, while the summer solstice angle reaches 74.0º.

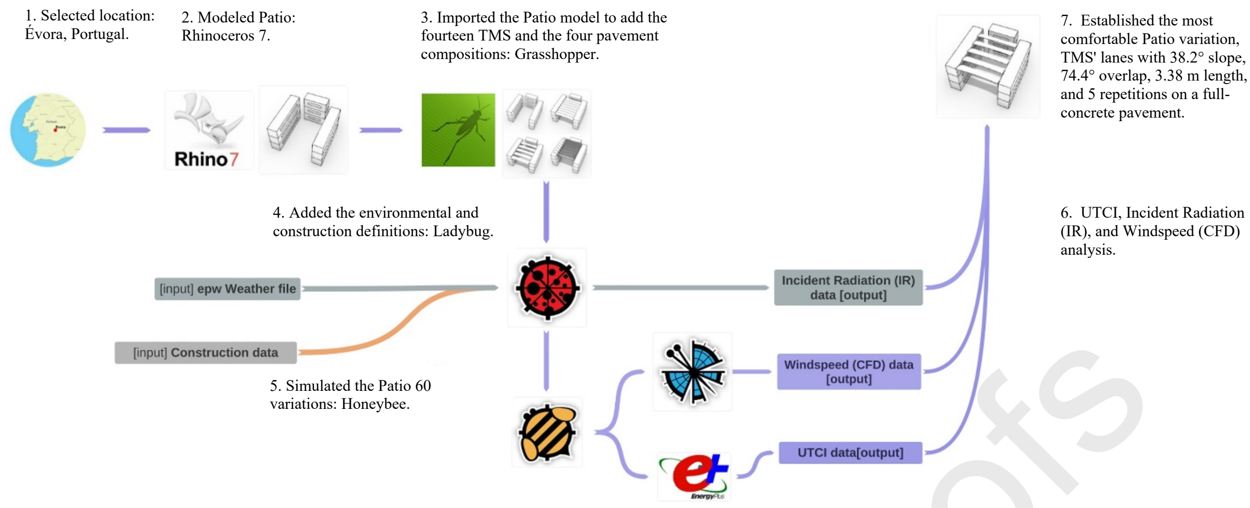
Today, the region is grappling with depopulation, emphasizing the need to attract newcomers, expatriates, migrants, and digital nomads. With a population of 53,591 individuals, the region faces a shortage of highly skilled professionals. Therefore, attracting an international workforce engaged in the high-tech industry, particularly in aircraft engineering and manufacturing, is crucial, as indicated by the 2021 Census survey [6].

TMS are popular due to their ability to withstand various environmental conditions, such as rain, infrared and ultraviolet radiation, excessive heat, acoustic disturbances, and lighting. TMS offer excellent reflectivity and minimal sunlight absorption, resulting in a significant decrease in both solar and heat intake within the structure. The membrane allows the passage of light, making it a suitable choice for regions characterized by warm or hot climates with high solar radiation values [7]. Moreover, these structures can actively modify the existing microclimate by providing shadow, affecting wind patterns, or even being able to adapt to specific climatic conditions [8].

These structures provide mechanical stability, light transmission, and environmental resilience, making them highly sought-after [9]. Designers have become aware of the precise fabrication, ease of assembly, competitive cost, environmental friendliness, and overall sustainability of TMS. However, reservations about designing and evaluating these structures persist, particularly regarding form finding, performance analysis, and identifying potential patterns [10]. This is where software becomes crucial, allowing for the precise assessment of these structures' shadow capabilities under energy simulation engines.

Considering this, the research team evaluated the performance of fourteen sets of TMS and four pavement surfaces (PS), as detailed in Tables 1 and 2, respectively (see Figure 1). The study focused on outdoor environmental factors, including temperature, relative humidity, solar radiation, and wind speed, within a semi-open U-shaped outdoor area (Patio) enclosed by maritime containers. The architectural solution reflects the traditional housing configuration of southern European countries. Following this goal, the study utilized EnergyPlus, a dynamic energy simulation engine, to assess the effectiveness of each TMS set compared to the uncovered Patio scenario.

Figure 1 — Research flowchart



The proposed geometrical model follows southern Europe's traditional housing patio structure, where central outdoor areas serve as places for socialization [11]. Nonetheless, the designed model adheres to the principles of rapid and straightforward assembly of lightweight structures, guaranteeing flexibility and adaptability to address regional emergent and critical situations [12]. These criteria align with TMS's offsite manufacturing and onsite assembly [13], [14]. However, none of these factors fully address the optimal user comfort from an outdoor environmental standpoint, as evaluated by the Universal Thermal Climate Index (UTCI).

The UTCI is a bioclimatic parameter that evaluates the physiological comfort of individuals based on specific weather conditions. It assesses the human body's response to the thermal environment following the air temperature and other factors such as relative humidity, windspeed, and radiation [15]. These conditions greatly influence our physiological reactions to the surrounding environment, which is crucial to ensure the pedestrians' capacity to work and relax in outdoor spaces [16].

According to the ANSI/ASHRAE Standard 55-2020, thermal comfort describes the state of mind that reflects satisfaction with the thermal environment and is evaluated subjectively. This concept, also referred to as human comfort, pertains to the level of satisfaction that occupants experience with the prevailing thermal conditions in a given space. When designing a structure intended for human occupation, it is crucial to consider the aspect of thermal comfort, ensuring that occupants are provided with an environment that meets their comfort needs [17].

2. Literature review

This section aims to show lightweight architectural models that meet standard comfort requirements in outdoor spaces. Its main goal is within the potential of lightweight architectural structures to ensure outdoor thermal comfort (OTC), which the UTCI assesses.

The importance of studying the thermal comfort levels in outdoor spaces is directly related to its impacts on the health and well-being of the population [18]. However, despite the broad research in this domain and its significance in recent years [19] [20] [21] [22] [23], it is worth mentioning the absence of studies that cross the potential of lightweight structures, such as the TMS, with OTC assessment by UTCI or other indexes. The few exceptions to this do not address South European or Mediterranean regions but rather tropical or hot-dry climate regions, which highlights the significance of the present study and its potential for broader applicability.

The literature on outdoor comfort and thermal climate indexes with membrane shading systems is still evolving, focusing on the impact of climate change and the quality of life in urban environments [24]. Shading systems, particularly vertically integrated ones, have been found to significantly enhance outdoor comfort and reduce the urban heat island effect [25]. However, the optimal urban system for hot and humid locations, improving outdoor thermal comfort, is still under debate [26]. The form and location of apertures in shading devices are key factors in achieving thermal comfort on an urban scale [27]. Further research is needed to develop a comprehensive understanding of the role of membrane shading systems in optimizing outdoor comfort under a Universal Thermal Climate Index.

The study of Elnokaly [28] under hot, arid regions investigates several conical TMS configurations used to predict the airflow behavior in enclosed and semi-enclosed spaces and its impact on user thermal comfort. The author conducted sixteen experiments to analyze real-scale and 1/20 scale models in 2D and 3D formats using computational fluid dynamics (CFD) software. The findings show, besides the reliability and predictability potential of computational techniques, that the influence of TMS geometry, shape, and orientation, along with other passive cooling measures, were shaping microclimate conditions that positively affect users' thermal comfort primarily by promoting evaporative cooling and increasing the air velocity around the human body. Also related, Ayoub and Ragheb [29] assessed the thermal behavior and potential applications of multiple tensile membrane roof designs, explicitly focusing on paraboloids, hyperboloids, and barrels in the harsh-hot desert climate of Aswan, Egypt. They utilized Grasshopper 3D and Ladybug Tools (LB) software for energy analysis. After generating 21,600 tensile membrane roofs (TMR), models were subjected to solar simulations and heat transfer analysis using the LB plugin combined with a specially developed Python code. The investigations involved a staggering 482,892,032 computational runs. The study provided a comprehensive understanding of the complex interactions of TMR in hot-desert climates. It identified 25 independent internal (geometry-related) and external (weather-related) variables that influence the performance of TMR. This research aimed to fill the gap in understanding TMR's heat transfer mechanisms and contribute to the broader field of sustainable construction practices. Like our study, their objective was to determine the optimal performance-form configuration and provide comprehensive data on tensile membrane performance. Ultimately, the authors suggested incorporating additional computational analyses, such as CFD, and using a robust climate dataset. There are methodological similarities between both studies and the one presented here. However, our approach does not focus on conical or paraboloid shapes but instead on a set of overlapping laminae. Moreover, addressing the recommendation by Ayoub and Ragheb [29], we consider airflow and wind speed and utilize a local EPW file with the most recent climate data to simulate a real-world scenario, thus increasing complexity. Lastly, our study aims to frame and optimize the results obtained within a comfort index (UTCI), a principle not considered by those mentioned above.

Regarding OTC, it is worth mentioning the study by Nouri et al.[30], conducted in Rossio Square in Lisbon, Portugal, where the Physiologically Equivalent Temperature (PET) assesses thermal loads and temporal worst-case scenarios without considering adaptive measures. The methodology employed CFD and Shadow Behavior Simulation studies, collecting data from six points of interest within the square, further processed by a comparative analysis reliant on Matzarakis, Mayer, and Iziomon chart [31]. The findings show that physiological stress reached "Extreme Heat Stress" levels, ranging from 51 to 56ºC. However, when shadowing measures were implemented, the recorded temperatures could be reduced by 16ºC, improving the thermal comfort in the square. The combination of urban complexity, limited space, poor ventilation, and global warming often creates heat islands that cause stress and make public urban spaces uncomfortable. A recent survey using thermal infrared cameras mounted on drones to perform aerial thermography examined the average radiant temperature of pedestrian urban spaces in Huelva, Spain, during a typical summer day. In conjunction with a microclimate simulation carried out on EM v5, this survey revealed that temperatures fluctuate by as much as 35ºC between 11 am and 3 pm, reaching a maximum of 59ºC and causing discomfort to users [32]. These findings highlight the importance of monitoring the impact of temperature and solar radiation on urban surfaces. Our study works similarly by adopting shadowing measures by exploring different TMS configurations to dampen the outdoor temperatures and keep them within a comfortable UTCI range.

In a more recent study in Baghdad, Iraq, Abbas and Khalid [33] intend to improve urban OTC by developing a "sustainable urban solution" prototype combining shading devices, high albedo surfaces, vegetation, and water features in a 16 by 16 m grid further assessed by the ENVI-met software to measure OTC in PET index. The findings were promising, showing that the prototype had a significant role in reshaping the microclimate of the urban configuration by improving the OTC to support social gatherings. Our study reached similar conclusions, although focused on the TMS impact on the OTC side rather than exploring the incorporation of additional elements such as vegetation. In this way, we could focus on the TMS dimensional, angular, and positional optimization adapted to the local climate conditions of Évora.

Still in the hot and dry climates, but in Semnan, Iran, Jafarian et al. [34] conducted a study to explore the potential of portable, lightweight membrane structures in the form of four canopy models. These models were designed to generate shading and natural ventilation patterns for creating an acceptable OTC. The authors used Ansys, ENVI-met, and LB plugins to simulate the outdoor temperatures. Their findings reveal a temperature difference of nearly 8ºC in this type of climate when using the membrane coverings, with the saddle-type canopy showing the most promising outcomes. In our assessment, we followed a similar methodology. However, instead of testing isolated portable canopies, our TMS is permanent and semi-enclosed in fourteen configurations. Furthermore, we evaluated the outcomes using UTCI, a factor not addressed by the authors of this study.

As part of sustainable urban development, outdoor microclimate simulations have significantly reduced energy consumption, CO2 emissions, urban heat islands, and urban sprawl. As stated by Elwy et al. [35] research, developing microclimate simulation tools is crucial to catch up with cutting-edge technologies and generate optimized urban forms according to their environmental performance. In their paper, a hybrid parametric workflow of LB validates their generic outdoor thermal comfort, calculated in Physiological Equivalent Temperature (PET), against measured meteorological data using HOBO-U30 weather station and EM simulations for a case study in Cairo, Egypt. The comparison of four meteorological parameters has shown minor error percentages with nearly 90% precise PET results. The study has demonstrated that LB has no less significance than other fully integrated microclimate simulation engines, particularly for consuming less simulation time and their appropriateness for the iterated design processes and simulations.

Lightweight membrane structures were evaluated not only in hot and dry climates. The study of Sultana and Bari [36] in the tropical climate of Gulshan, Dhaka, Bangladesh, covered two public spaces using TMS configurations and assessed their comfort level using predicted mean vote (PMV) and predicted percentage dissatisfied (PPD) models. The findings revealed that using TMS can offer a viable solution to improve OTC by reducing temperatures beneath it in tropical climates. Lastly, Chen et al. [37] evaluate the influence on OTC of different pavement surfaces and shadowing devices (vegetation and TMS) in high altitudes, specifically in Lhasa City, Tibet. The authors resorted to measurements in situ and statistical analysis to process data, further applying the PET index to assess OTC. The findings show that areas adorned with deciduous trees enhance OTC more effectively than TMS, showcasing a temperature difference of up to 9.73ºC during the summer. Concerning surface materials, lawns exhibited the best results mainly because they generate a favorable microclimate. Despite the validity of the OTC assessment outcomes, both studies do not delve into exploring an optimal TMS configuration, leaving room for further research and development. In contrast, one of the core objectives of our study is to precisely explore the optimal configuration to achieve the best OTC annual outcomes while providing an in-depth explanation of the process. Moreover, both studies do not focus on designing an architectural solution that accounts for pavement surfaces and shadow devices intended to be built in deserted areas, but rather, they assess spaces that have already been constructed.

Provided this, the study intends to (1) fill the research gap by evaluating the physiological comfort of individuals based on specific weather conditions, as assessed by the UTCI, and; (2) analyze the impact of shadowing on occupant comfort as measured by the UTCI. The research gap is in the absence of studies that cross the potential of lightweight structures, such as TMS, with outdoor thermal comfort assessment by the UTCI or other indexes, particularly in South European regions. Moreover, published research does not address analysis that seeks to optimize TMS performance for outdoor comfort based on bioclimatic parameters such as the UTCI.

The study also aims to ensure that the Patio's UTCI change in temperature (ΔT) remains within the minimum comfort gap. During winter, it allows for the maximum solar radiation. During summer, it provides shade while ensuring adequate windspeed for the comfort of Patio users. This comprehensive analysis and optimization of TMS configurations for outdoor comfort based on bioclimatic parameters such as the UTCI represent the identified research gap in the provided attachment.

3. Materials and Methods

3.1. Architectural approach

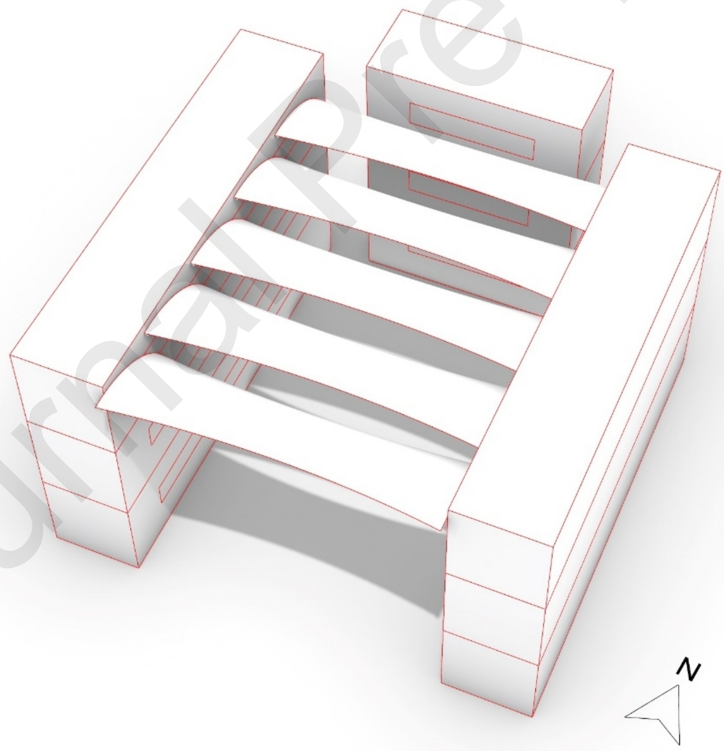
The success of people's interactions depends on the comfort of external spaces, which typically leads to higher population dynamics and integration [32]. In this regard, the architectural design of a semi-open U-shaped Patio aims to create a micro-environment that enhances users' comfort by maintaining optimal temperature, relative humidity, and solar radiation levels, as well as reducing the impact of prevailing winds, particularly the north-south oriented during summer, which are known for their strong, chilly gusts from the north, locally referred to as nortada.

The Patio features various pavement surfaces, including concrete and a turf bed, in varying percentages. These combinations help to maintain a comfortable temperature due to a higher thermal mass delay and favor a high relative humidity, cutting the air temperature during peak solar radiation hours of warmer seasons, particularly in spring, summer, and fall. The high thermal mass delay also absorbs energy from the surrounding temperature and direct solar radiation, mitigating extremely low temperatures. The open area of the Patio faces south and it is covered with a membrane to maximize winter sun radiation and summer shading, which reduces high temperatures and smooths out heat waves in inland Mediterranean climates. The membrane also deflects high north-south wind speeds during both seasons. When all these factors are balanced, it creates an optimal outdoor environment for human comfort assessed by the UTCI, or user thermal sensation.

3.2. Phases Description

The study uses EnergyPlus 2022, an energy simulation engine integrated with LB 1.5, an open-source Python software, to perform the environmental analysis and simulation for building and outdoor design upon Grasshopper, a visual programming environment for Rhinoceros 7[3]. The simulation aims to analyze the shadow performance of the TMS under various arrangements, slopes, lengths, and overlap angles, which define the membrane lanes (repetitions), to gather Patio UTCI. All the simulations follow an annual assessment (January 1st to December 31st). The assessment follows Figure 2 and Tables 1, 2, and 3. The evaluation begins with modeling a set of containers in Rhinoceros 7, which includes nine maritime containers arranged in a three-level U-shaped Patio design. The "U opposite arms" consist of six “40 feet” type general purpose containers with 12.19m length, 2.44m width, and 2.59m height. In comparison, the "U connection" uses three “20 feet” type general purpose container with 6.05m length, 2.44m width, and 2.59m height, resulting in a Patio with an interior dimension of 9.05m width by 12.19m length and 7.77m height. A TMS covers the Patio with different surface arrangements, as shown in Figure 2.

Figure 2 —Patio model



The Patio simulation model assesses the suburban region of Évora, with ground vegetation, several medium-sized trees measuring 5 to 10m, and mostly olive trees on rock wall plots. The standard container surfaces consist of corrugated painted steel sheets with a thickness of 1.5mm

[3] This software can also integrate with other libraries and frameworks, such as NumPy, SciPy, and OpenStudio

and a reinforced structural frame. The entire structure sits on a reinforced concrete slab with four layers, including a waterproof PVC membrane, sand, a geotextile membrane, and various sizes of rockfill, all inserted into dug and compacted soil. The membrane systems intersect with four different Patio pavement surfaces to enhance the Patio's UTCI performance, as shown in Table 1.

The tensile membrane used in the project is a white-colored polytetrafluoroethylene (PTFE) fiberglass fabric coated with Teflon® that ensures 100% non-combustibility and a Class "A" rating under sunlight exposure. With 73% solar reflectivity and 87% visible reflectivity/13% visible light transmittance (VLT), the membrane has a thickness of 55µm and a life expectancy of 30 years, according to the ASTM E424 norm.

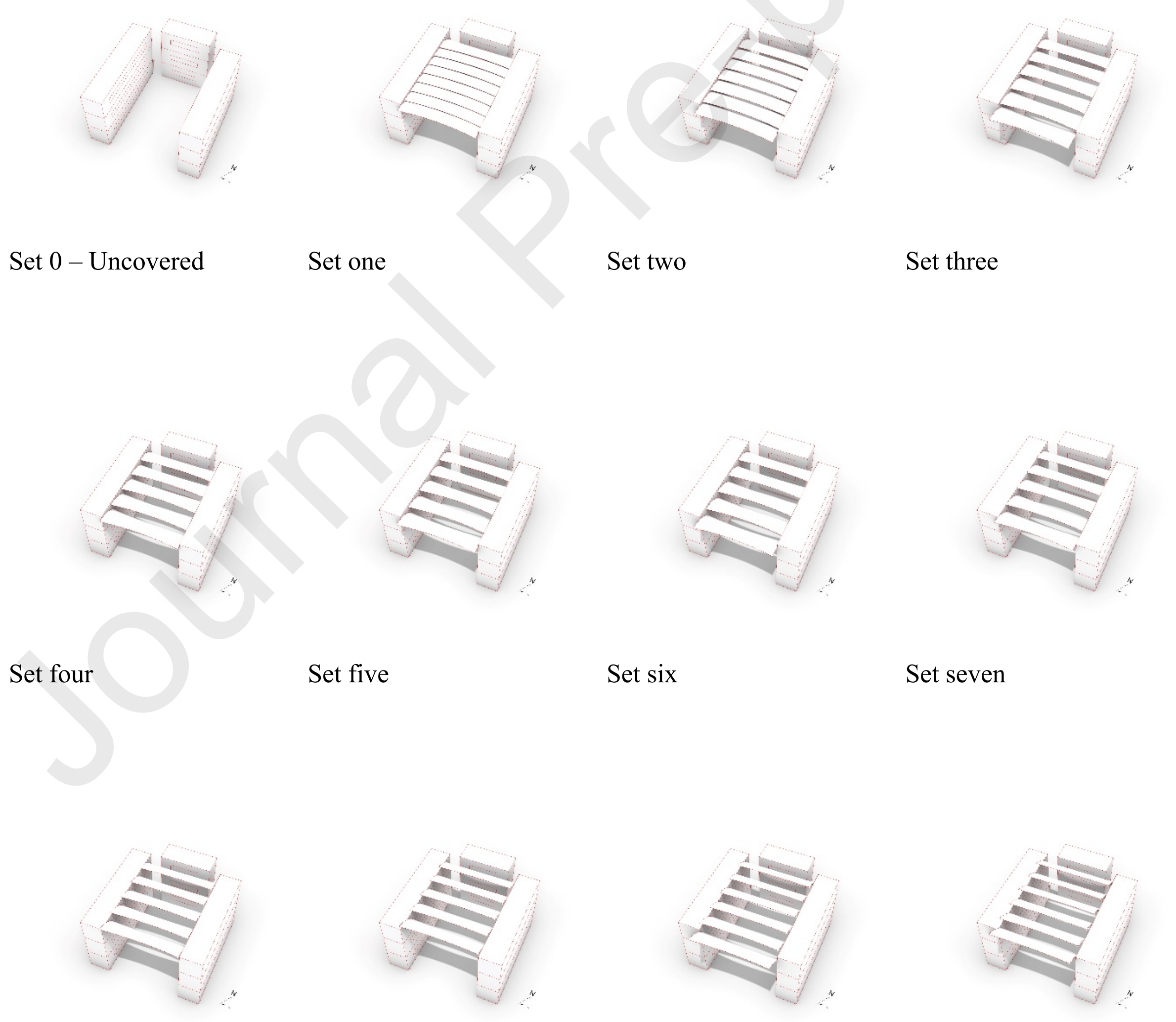
Table 1 — Membrane angles, length, and repetitions.

Patio set	Configuration date (Slope/overlap)	Slope	Overlap	Length (m)	Repetitions
Uncovered	non applies				
Covered					
One	Winter solstice/March 5th	27.6º	44.1º	3.44	8
Two	Winter solstice/Spring equinox	27.6º	50.2º	3.44	7
Three	Winter solstice/April 20th	27.6º	62.4º	3.44	6
Four	Winter solstice/May 6th	27.6º	67.6º	3.44	5
Five	Winter solstice/May 20th	27.6º	71.0º	3.44	5
Six	Winter solstice/ Summer solstice	27.6º	74.4º	3.44	5
Seven	January 31st/ Summer solstice	33.2º	74.4º	3.64	5
Eight	February 16th/ Summer solstice	38.2º	74.4º	3.88	5
Nine	February 23rd/ Summer solstice	40.7º	74.4º	4.02	5
Ten	March 2nd/ Summer solstice	43.3º	74.3º	4.19	6
Eleven	March 10th/ Summer solstice	46.4º	74.3º	4.42	6

Twelve	Spring equinox/ Summer solstice	50.4º	74.3º	4.79	6
Thirteen	April 21st/ Summer solstice	62.6º	74.2º	6.63	8
Fourteen	May 5th/ Summer solstice	67.0º	74.1º	7.79	11

Table 1 presents the various configurations obtained in the study, starting from set number one, corresponding to the winter solstice with the highest closed overlap angle (March 5th). As the study progresses to set six, the overlap angles open to reach the summer solstice angle, covering the more exposed configurations. From sets six to twelve, as illustrated in Figure 3, the overlap aligns with the summer solstice angle, and the slope increases gradually from January 31st to the spring equinox angle, exposing the surface to more radiation.

Figure 3 — TMS models from set zero to fourteen.



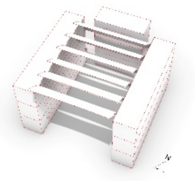
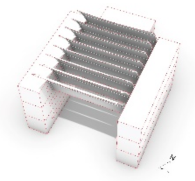
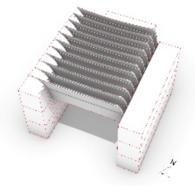
Set eight	Set nine	Set ten	Set eleven
			
Set twelve	Set thirteen	Set fourteen	

Table 2 presents various pavement sets, material characteristics, and properties from the EnergyPlus database. These parameters are intended for the subsequent simulation of the TMS to determine the best-performing Patio. The introduction of grassy lawns aims to balance relative humidity levels and temperatures. The inclusion of concrete pavement is intended to act as an accumulator to balance and delay thermal heat fluctuations.

Table 2— Patio pavement sets, characteristics, and properties.

	Emissivity	**Heat Capacity**	**Density**	**Albedo**
PS#1 (100% Grassy lawn turf bed)	0.95 ε	1200 J/Kg-k	1100 kg/m³	0.22
PS#2 (36% Concrete and 64% Grassy lawn turf bed)	0.64 ε	1069 J/Kg-k	1512 kg/m³	0.27
PS#3 (67% Concrete and 33% Grassy lawn turf bed)	0.38 ε	957 J/Kg-k	1866 kg/m³	0.31
PS#4 (100% Concrete)	0.1 ε	837 J/Kg-k	2243 kg/m³	0.35

3.2.1. Phase II – Simulation inputs

The study proceeds with a systematic comparative analysis of fourteen covered Patio sets (1-14), with the uncovered Patio as the baseline for comparison. The TMS in each set is combined with four pavement surfaces to calculate the monthly average UTCI ΔT, with all readings based on EnergyPlus 2022 calculations using Évora's EPW file[4].

[4] ISO 9920 (2007), Ergonomics of the thermal environmental — Estimating a clothing ensemble's thermal insulation and water vapor resistance. International for Standardization, Geneve.

The team selected the UTCI to evaluate the temperature experienced by the human body, which considers the mean radiant temperature (MRT), air temperature, solar radiation, wind speed of 0.5 m/s at 10 m, relative humidity of 50%, water vapor pressure of 20 hPa, and a metabolic rate of 135W/m2.

The aim is to simulate the Patio's UTCI from 9 am to 5 pm, tracking the path of sunlight based on latitude and evaluating the performance of the TMS to produce shadow, retain heat, and manage moisture, in addition to the different pavement surfaces to balance thermal fluctuations.

UTCI is applicable across various climates and seasons, assuming that individuals are walking with a metabolic rate of approximately 2.4 met and adapt their clothing according to the outdoor temperature. LB recommends the PET model when these specifications are not met [38]. The clothing behavior of individuals, specifically the amount of insulation they choose to wear, is influenced by the air temperature in a non-linear manner within the range of 0.4 clo to 3.0 clo [39]. The UTCI calculations require four variables: air temperature, dew point temperature (or relative humidity), both at 2m above ground level, wind speed at 10m above ground level, and MRT [40], as shown in Table 3.

In the Mediterranean region, the UTCI comfort range is typically between 20 and 25ºC, with a wind speed of 0.2 m/s [41], representing the optimal comfort level for outdoor environments. The team chose to use different reference conditions, in line with the UTCI ΔT calculation published by Action Cost 730 [42], which defines low user activity, such as walking at 4 km/h with a metabolic rate of 135W/m2 and mild climatic conditions where the radiant temperatures are equal to the ambient temperature, with a wind speed of 0.5 m/s at the height of 10 m, a relative humidity of 50% for Ta<29ºC, and an atmospheric pressure of 2 kPa for Ta>29ºC, as indicated in Table 3 [43].

Table 3 — UTCI, reference conditions [43], [38], [39], [40][5]

	Reference	Condition
Activity	Walking	4km/h (135W/m^2)
Clothing	Adaptative	0.4 clo to 3.0 clo
Climate	Radiant Temperature and Ambient Temperature (T_r=T_a)	$V_{a,\ 10m}$ = 0.5m/s or 0.3m/s at 1.1m heigh
	Relative humidity, rH = 50%	TA<29ºC
	Atmospheric Pressure, P_a = 2kPa	TA>29ºC
Calculation conditions	Air temperature (ºC)	2m above ground level

[5] ISO 9920 (2007), Ergonomics of the thermal environmental — Estimating a clothing ensemble's thermal insulation and water vapor resistance. International for Standardization, Geneve.

	Mean Radiant Temperature (ºC)	-
	Relative humidity (%)	2m above ground level
	Windspeed (m/s)	10m above ground level

3.2.2. Phase 3 – Data organization

The team ran 720 year-based simulations to identify the best whole model configuration. The analysis began by extracting results on the lowest deviation from the comfort gap range set (>9ºC to <26ºC) by accessing the Thermal Comfort Percent (TCP), Heat Sensation Percent (HSP), and Cold Sensation Percent (CSP).

The study utilized LB Honeybee and Butterfly suite component readings to analyze various TMS arrangements in combination with different Patio pavement finishes (PS#1 – 4, see Table 2). Additionally, the team conducted 84 dedicated simulations to assess the UTCI breakdown of the best-performing solution, considering air temperature, relative humidity, water vapor pressure, incident radiation, and wind speed.

Table 4 outlines the inputs used in LB tools for the simulations, which were then processed by the EnergyPlus engine, as listed in Table 1.

Table 4 — EnergyPlus simulation inputs

Paraments	Values
Number of zones (exterior)	1 — 9.05m (width) x 12.19m (length) x 7.77m (hight)
Solar orientation	Patio open area = azimuth 0º
Simulation period	9 am to 5 pm (January 1st to December 31st)
Time interval	One timestep/hour
Reference location	Évora, Portugal (38° 34' 28.008" E; -7° 54' 27.559" N)
Distance from shore	80km
Altitude	265m
Weather file	PRT_Evora.085570_IWEC\PRT_Evora.085570_IWEC.epw
Zone occupancy	0.33 persons/m^2
Sensible metabolic heat gain [W/person]	70

Latent metabolic heat gain [W/person]	50
Ground reflectance	0.20
Wind exposition	Following the EPW data
Horizontal shadowing	Limited to TMS plus containers

Materials and components [45]	**Emissivity**	**Heat capacity**	**Density**	**Albedo**
Tensile Membrane (white-colored PTFE)	0.92ε	1000 J/kgK	2200 kg/m^3	0.85
Containers surface	0.88 ε	450 J/kgK	7800 kg/m^3	0.2
Patio surface (corrugated painted steel sheets of red oxide color/ grassy lawn turf bed)	0.85ε / 0.98ε	1000 J/kgK / 3000 J/kgK	2200 kg/m^3 / 1050 kg/m^3	0.2-0.3 / 0.25[6]

The LB Butterfly CFD simulation engine operates under the following conditions: see Table 5.

Table 5 — LB Butterfly CFD

Components	**Values**
Wind tunnel	_windward_x_ : 5 _top_x_ : 5 _sides_x_ : 5 _leeward_x_ : 10
Create a case from a tunnel	_ref_wind_height: 1 m
Mesh	_grad_xyz_ : simpleGrading (1,1,1) (default) _cell_count_: 5 (default)
Mesh refinement_snappyHexMesh_	_glob_refine_level_ : (6,6)

A system of 96 virtual sensors captures all data placed one meter above the Patio pavement. The system lies on a 2D mesh where a meter grid organizes the sensors. The figures collected by the virtual data loggers build the average temperatures presented in the results section: Tables 7, 8, 9, and 10.

[6] https://decrolux.com.au/learning-centre/2018/approximate-reflectance-values-of-typical-building-finishes

4. Results

This section presents the results of the annual UTCI readings from 60 variations to identify the best-performing Patio configuration. A lower UTCI ΔT value signifies higher user comfort, with beneficial winter solar radiation and enhanced summer protection.

The tables below show the performance of various sets according to UTCI ΔT readings, represented by a color code (refer to Table 3). The color code indicates "slightly cold stress" (blue) for 0ºC to +9ºC, "no thermal stress" (green) for +9ºC to +26ºC, and "moderate heat stress" (orange) for above +26ºC to +32ºC (see Table 6).

Table 6 — Colors codes of tables 7, 8, 9, and 10 [42].

Color	Temperature range – UTCI		Condition
	0ºC	**<9ºC**	**Slight cold stress**
	≥9ºC	**≤26ºC**	**No thermal stress**
	>26ºC	**32ºC**	**Moderate heat stress**

Tables 7, 8, 9, and 10 present the monthly comfort gaps from the simulations, enabling direct comparisons between sets using color-coded UTCI readings, as specified in Table 6. The "Total" value in the tables represents the sum of temperatures below 9ºC and above 26ºC, indicating the underperformance range of each set.

Table 7 features the uncovered Patio with a full grassy lawn turf bed (PS#1) and shows the most uncomfortable condition with a UTCI ΔT = 12.0ºC. For temperatures below 9ºC, the ΔT = 3.3ºC and above 26ºC, ΔT = 8.7ºC, which all other solutions surpass, with the best performances in sets two and three. These sets share the same slope and length but have different overlapping angles (50.2º and 62.4º, respectively). The advantage of these lower angles is evident in January and December, with a ΔT = 4.1ºC (<9ºC) and reduced deviations in other months. Sets one, two, and three, with 0ºC (>26ºC), benefit from their low slope angles (27.6º), overlap angles (41.4º for set 1, 50.2º for set 2, and 62.4º for set 3), and identical length of 3.44m. However, the eight and seven repetitions in sets one and two, respectively, are less effective compared to the six repetitions in set three, which optimizes the TMS's overall costs. Set three (highlighted in bold) stands out for its superior performance, reducing the total ΔT by 7.9ºC compared to the uncovered Patio, from ΔT = 12.0ºC to ΔT = 4.1ºC.

Table 7 — Patio monthly UTCI (ºC) simulations with PS#1.

	UTCI (ºC)												Comfort gap ΔT (ºC)		
Patio set	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	<9	> 26	Total

Uncovered	6.6	11.5	14.4	15.4	22.9	26.2	29.9	29.5	27.1	20.3	11.0	8.1	3.3	8.7	12.0
Covered															
One	5.5	9.4	11.1	10.8	17.7	21.1	25.2	25.3	23.8	18.0	9.7	7.3	4.2	0.0	4.2
Two	5.6	9.5	11.2	10.8	17.6	21.0	25.1	25.2	23.8	18.2	9.7	7.3	4.1	0.0	4.1
Three	**5.6**	**9.7**	**11.6**	**11.1**	**17.8**	**21.2**	**25.3**	**25.6**	**24.2**	**18.3**	**9.8**	**7.3**	**4.1**	**0.0**	**4.1**
Four	5.8	10.0	12.1	11.7	18.5	21.7	25.9	26.2	24.6	18.7	10.0	7.5	4.7	0.2	4.9
Five	5.8	10.0	12.1	11.7	18.5	21.7	25.9	26.2	24.6	18.7	10.0	7.5	4.7	0.2	4.9
Six	5.8	10.0	12.0	11.7	18.5	21.8	25.9	26.2	24.6	18.6	10.0	7.4	4.8	0.2	5.0
Seven	5.8	10.1	12.3	11.9	18.7	21.8	26.0	26.4	24.8	18.8	10.1	7.5	4.7	0.4	5.1
Eight	5.9	10.2	12.5	12.1	18.8	21.9	26.1	26.5	25.0	19.0	10.1	7.5	4.6	0.6	5.2
Nine	5.9	10.3	12.7	12.2	18.9	22.0	26.2	26.6	25.1	19.1	10.1	7.6	4.5	0.8	5.3
Ten	5.5	10.0	12.5	11.9	18.5	21.6	25.8	26.3	24.9	18.8	9.8	7.2	5.3	0.3	5.6
Eleven	5.6	9.9	12.6	12.1	18.7	21.7	25.9	26.5	25.0	18.9	9.8	7.3	5.1	0.5	5.6
Twelve	5.6	9.9	12.7	12.3	18.9	22.0	26.2	26.8	25.2	18.8	9.8	7.4	5.0	1.0	6.0
Thirteen	5.6	9.7	11.8	12.4	19.3	22.1	26.4	27.2	24.8	18.4	9.8	7.4	5.0	1.6	6.6
Fourteen	5.3	9.4	11.5	11.8	19.1	22.0	26.3	26.7	24.3	18.2	9.6	7.2	5.5	1.0	6.5

Table 8 reveals that the uncovered Patio exhibits the most comfortable condition with a UTCI ΔT = 11.4ºC, an improvement over the same TMS configuration in Table 7. Introducing thermal mass (PS#2) on the Patio pavement reduces the temperature variation from ΔT = 3.3ºC (PS#1) in Table 7 to ΔT = 3.1ºC (PS#2) in Table 8 for temperatures under 9ºC. Above 26ºC, the uncovered set's temperature variation decreases from ΔT = 8.7ºC (PS#1) to ΔT = 8.3ºC (PS#2), outperforming every other solution. This configuration leads to better performance in the ninth set, featuring a slope angle of 40.7º, an overlap of 74.4º, a length of 4.02 m, and five repetitions. Higher angles contribute to reduced radiation during the warmer summer months, decreasing thermal mass delay. Eight sets (1, 2, 3, 4, 5, 6, 7, 10) show a 0ºC (>26ºC) due to lower slope angles or significant overlapping. The repetitions favor sets four to six, each with five repetitions, reducing the overall costs. Among these compositions, sets four to six perform similarly, but from the perspective of dynamic wind charges, lower slopes are preferred, with the fourth set standing out (shown in bold).

Table 8 — Patio monthly UTCI (º C) simulations with PS#2.

	UTCI (ºC)												Comfort gap ΔT (ºC)		
Patio set	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	<9	> 26	Total
Uncovered	6.7	11.6	14.5	15.4	22.8	26.1	29.8	29.4	27.0	20.3	11.1	8.2	3.1	8.3	11.4
Covered															
One	5.6	9.5	10.9	10.7	17.3	20.7	24.8	25.0	23.6	18.0	9.7	7.4	5.0	0	5.0
Two	5.6	9.6	11.1	10.6	17.2	20.6	24.6	24.9	23.6	18.1	9.8	7.4	5.0	0	5.0
Three	5.7	9.7	11.4	11.0	17.5	20.8	24.8	25.2	24.0	18.3	9.9	7.5	4.8	0	4.8
Four	**5.9**	**10.1**	**12.0**	**11.6**	**18.2**	**21.4**	**25.5**	**25.9**	**24.5**	**18.7**	**10.1**	**7.6**	**4.5**	**0**	**4.5**
Five	5.9	10.0	11.9	11.6	18.2	21.4	25.5	25.9	24.5	18.6	10.1	7.6	4.5	0	4.5
Six	5.9	10.0	11.9	11.6	18.3	21.5	25.5	25.9	24.5	18.6	10.1	7.6	4.5	0	4.5
Seven	6.0	10.2	12.2	11.8	18.4	21.5	25.6	26.1	24.7	18.8	10.2	7.6	4.4	0	4.4
Eight	6.0	10.3	12.5	11.9	18.5	21.6	25.7	26.2	24.9	19.0	10.2	7.7	4.3	0.2	4.5
Nine	6.1	10.4	12.6	12.1	18.6	21.7	25.8	26.3	25.0	19.1	10.2	7.7	4.2	0.3	4.5
Ten	5.6	10.0	12.4	11.8	18.2	21.3	25.4	26.0	24.8	18.8	9.9	7.4	5.0	0	5.0
Eleven	5.7	10.0	12.6	12.0	18.4	21.4	25.5	26.2	24.9	18.8	9.9	7.4	4.9	0.2	5.1
Twelve	5.7	10.0	12.7	12.2	18.7	21.7	25.8	26.5	25.1	18.8	10.0	7.5	4.8	0.5	5.3
Thirteen	5.7	9.8	11.8	12.3	19.0	21.8	26.0	26.9	24.7	18.4	9.9	7.5	4.8	0.9	5.7
Fourteen	5.4	9.4	11.3	11.7	18.8	21.7	25.9	26.4	24.2	18.2	9.6	7.3	5.3	0.4	5.7

Table 9 presents the uncovered set with the same global UTCI ΔT = 11.4ºC as in Table 8. Compared to the temperatures under 9ºC, ΔT is 3.1ºC (PS#2) in Table 8 and 2.8ºC (PS#3) in Table 9. Above 26ºC, the opposite trend is observed, with ΔT = 8.3ºC (PS#2) in Table 8 and ΔT = 8.6ºC in Table 9, both due to the inversion of pavement thermal masses. The ninth set (shown in bold) achieves the best performance due to the energy harvested and increased thermal mass area. Under 9ºC, the repetitions favor sets four to nine, each with five repetitions; however, the eighth set stands out at 3.9ºC (<9ºC), outperforming the others. Moreover, the eighth set (shown in bold) impedes the UTCI readings above 26ºC due to the precise shadow balance for Évora's latitude (38º). Additionally, it is valid for ten sets (1, 2, 3, 4, 5, 6, 7, 8, 10, 11) showing a 0ºC (>26ºC), favoring lower slope angles or significant overlapping.

Table 9 — Patio monthly UTCI (º C) simulations with PS#3.

	UTCI (ºC)												Comfort gap ΔT (ºC)		
Patio set	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	<9	> 26	Total
Uncovered	6.8	11.8	14.6	15.6	22.9	26.2	29.8	29.5	27.1	20.4	11.3	8.4	2.8	8.6	11.4
Covered															
One	5.8	9.6	10.8	10.6	17.0	20.4	24.3	24.6	23.5	18.0	9.9	7.6	4.6	0	4.6
Two	5.8	9.7	11.0	10.5	16.9	20.3	24.2	24.5	23.6	18.2	9.9	7.6	4.6	0	4.6
Three	5.8	9.9	11.5	10.9	17.2	20.5	24.4	24.9	24.0	18.4	10.0	7.6	4.6	0	4.6
Four	6.1	10.2	12.0	11.6	18.0	21.2	25.1	25.7	24.5	18.7	10.2	7.8	4.1	0	4.1
Five	6.1	10.2	12.0	11.6	18.0	21.1	25.1	25.7	24.5	18.7	10.2	7.8	4.1	0	4.1
Six	6.0	10.2	12.0	11.6	18.0	21.2	25.2	25.7	24.5	18.7	10.2	7.7	4.3	0	4.3
Seven	6.1	10.4	12.3	11.8	18.2	21.3	25.3	25.9	24.7	18.9	10.3	7.9	4.0	0	4.0
Eight	**6.2**	**10.5**	**12.6**	**12.0**	**18.3**	**21.3**	**25.4**	**26.0**	**24.9**	**19.1**	**10.4**	**7.9**	**3.9**	**0**	**3.9**
Nine	6.3	10.5	12.7	12.1	18.4	21.4	25.5	26.2	25.0	19.2	10.4	7.9	3.8	0.2	4.0
Ten	5.8	10.2	12.5	11.8	18.0	21.0	25.0	25.8	24.8	18.9	10.1	7.6	4.6	0	4.6

Eleven	5.9	10.2	12.7	12.0	18.2	21.1	25.2	26.0	25.0	19.0	10.2	7.6	4.5	0	4.5
Twelve	5.9	10.3	12.8	12.3	18.5	21.4	25.6	26.3	25.2	19.0	10.1	7.7	4.4	0.3	4.7
Thirteen	5.9	10.0	11.9	12.4	18.9	21.6	25.8	26.8	24.8	18.6	10.1	7.7	4.4	0.8	5.2
Fourteen	5.5	9.6	11.2	11.8	18.7	21.4	25.6	26.3	24.2	18.2	9.8	7.4	5.1	0.3	5.4

Table 10, the second highest due to the entire concrete Patio surface (PS#4) utilizing thermal mass delay, especially during summer, shows a UTCI ΔT = 9.5ºC. Under low temperatures of 9ºC, the temperature gap decreases from ΔT = 2.8ºC (PS#3) in Table 9 to ΔT = 2.3ºC (PS#4) in Table 10. The best-performing solutions are in the eighth (shown in bold) and ninth sets, with higher slopes that favor solar radiation in winter. The eighth set (shown in bold) UTCI underperforms in January and December, compared to the uncovered set, falling out of the comfort gap under the 9ºC mark by ΔT = 3.4ºC against ΔT = 2.3ºC but outperforming during the other months by surpassing the 26ºC mark by ΔT = 0.1ºC against ΔT = 9.5ºC, respectively. From a yearly perspective, they show a positive result of ΔT = 3.5ºC against ΔT = 11.8ºC of the uncovered Patio. Eight covered sets (1, 2, 3, 4, 5, 6, 7, 10, 11) show ΔT = 0ºC due to lower slope angles or significant overlapping from February to November.

The repetitions favor sets four to nine, all with five repetitions, reducing overall TMS costs. However, the eighth and ninth sets under PS#4 present slightly higher ΔT values of 0.1ºC and ΔT 0.2ºC (>26ºC), respectively. These sets achieve the best performance among the 720 simulations run, with an overall score of ΔT = 3.5ºC. Considering the overall costs and higher exposure to dynamic wind charges, the best TMS set is the eighth, with a slope of 38.2º, an overlap of 74.4º, a length of 3.88 m, and five repetitions.

Table 10 — Patio monthly UTCI (º C) simulations with PS#4.

	UTCI (ºC)												Comfort gap ΔT (ºC)		
Patio set	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	<9	> 26	Total
Uncovered	7.1	12.0	15.0	15.9	23.1	26.4	30.0	29.7	27.4	20.6	11.5	8.6	2.3	9.5	11.8
Covered															
One	6.0	9.9	11.1	10.7	16.9	20.1	24.0	24.6	23.6	18.2	10.1	7.8	4.2	0	4.2
Two	6.0	10.0	11.2	10.6	16.7	20.0	23.8	24.4	23.6	18.4	10.2	7.8	4.2	0	4.2
Three	6.1	10.1	11.7	11.0	17.0	20.2	24.1	24.8	24.0	18.5	10.2	7.9	4.0	0	4.0

Four	6.3	10.5	12.3	11.7	17.9	21.0	24.9	25.7	24.6	18.9	10.5	8.0	3.7	0	3.7
Five	6.3	10.4	12.3	11.7	17.9	20.9	24.9	25.6	24.6	18.9	10.5	8.0	3.7	0	3.7
Six	6.3	10.4	12.3	11.7	17.9	21.1	24.9	25.6	24.6	18.9	10.4	8.0	3.7	0	3.7
Seven	6.4	10.6	12.6	11.9	18.1	21.1	25.1	25.9	24.8	19.1	10.6	8.0	3.6	0	3.6
Eight	**6.4**	**10.8**	**12.9**	**12.2**	**18.2**	**21.2**	**25.2**	**26.1**	**25.1**	**19.3**	**10.6**	**8.2**	**3.4**	**0.1**	**3.5**
Nine	6.5	10.8	13.1	12.3	18.4	21.3	25.3	26.2	25.2	19.4	10.6	8.2	3.3	0.2	3.5
Ten	6.0	10.5	12.8	12.0	17.9	20.8	24.8	25.8	25.0	19.1	10.4	7.9	4.1	0	4.1
Eleven	6.3	10.6	13.1	12.2	18.1	20.9	25.0	26.0	25.2	19.2	10.5	7.9	3.8	0	3.8
Twelve	6.2	10.6	13.1	12.5	18.5	21.4	25.4	26.4	25.4	19.1	10.5	8.0	3.8	0.4	4.2
Thirteen	6.1	10.3	12.2	12.6	18.9	21.6	25.6	26.9	24.9	18.8	10.4	7.9	4.0	0.9	4.9
Fourteen	5.8	10.0	11.6	12.1	18.7	21.3	25.6	26.5	24.4	18.7	10.0	7.7	4.5	0.5	5.0

Figures 4 and 5 compare the total UTCI ΔT for all covered and uncovered sets from Table 10, with the green color highlighting the TMS best-performing set. The performance line for all months starts at ΔT = 11.8ºC for the uncovered set and rapidly declines to ΔT = 4.2ºC for one set due to shadowing. The line reaches its lowest point at sets eight and nine with ΔT = 3.5ºC. Higher slopes after set ten reduce the deviation under similar overlapping, necessitating increased repetitions.

Figure 4 —TMS set features with PS#4 (angles and overlaps) and outcomes (UTCI ΔT)

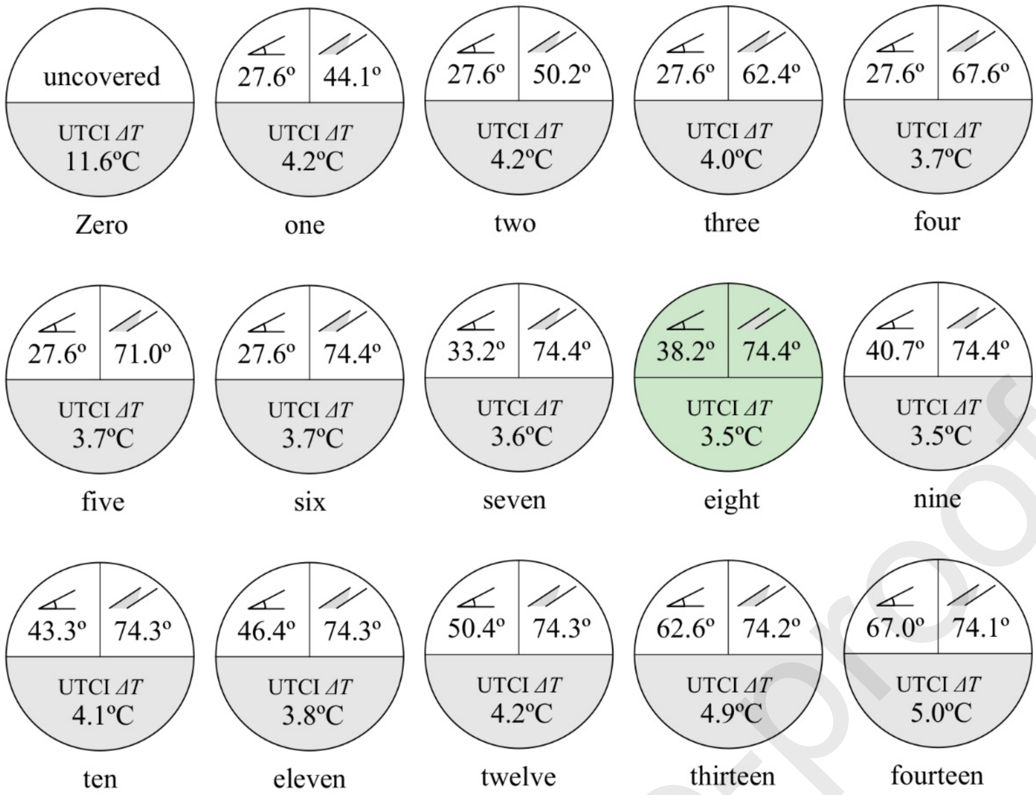


Figure 5 — Total UTCI ΔT (ºC) of Table 7 from set zero (uncovered) to TMS set fourteen.

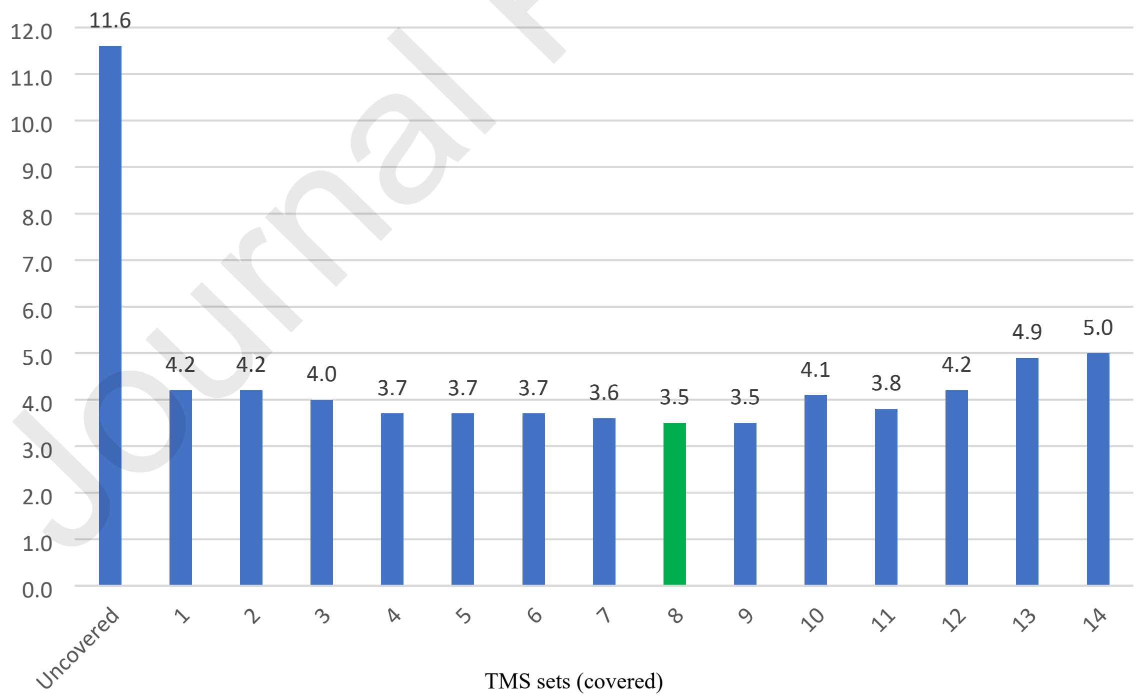


Table 11 displays the best-performing solution's TCP, HSP, and CSP values under all Patio pavement surfaces, as obtained by the LB Honeybee UTCI Comfort Map component. The data strongly support using the eighth set with the entire colorless concrete pavement surface, which yields the highest TCP of 47.7%, indicating the highest overall comfort. This is divided into an HSP of 7.4% and a CSP of 44.9%, suggesting discomfort on colder days and comfort on warmer ones. These results reflect the increased thermal mass delay on the Patio surface, particularly between 9ºC and 26ºC.

Table 11 — TCP, HSP, and CSP of the best-performance (24h).

Pavement thermal mass behavior	TCP (%)	HSP (%)	CSP (%)
PS#1 - Three set	47.2	7.8	46.0
PS#2 - Four set	46.7	7.5	45.8
PS#3 - Eighth set	47.2	7.4	45.4
PS#4 - Eighth set	47.7	7.4	44.9

The study gathers UTCI data by extracting information from the Évora weather file, which includes ambient temperature, relative humidity, water vapor pressure, incident radiation, and wind speed. The LB Honeybee Read Environment Matrix component correlates these parameters with the proposed Patio microenvironment to gather air temperature and relative humidity.

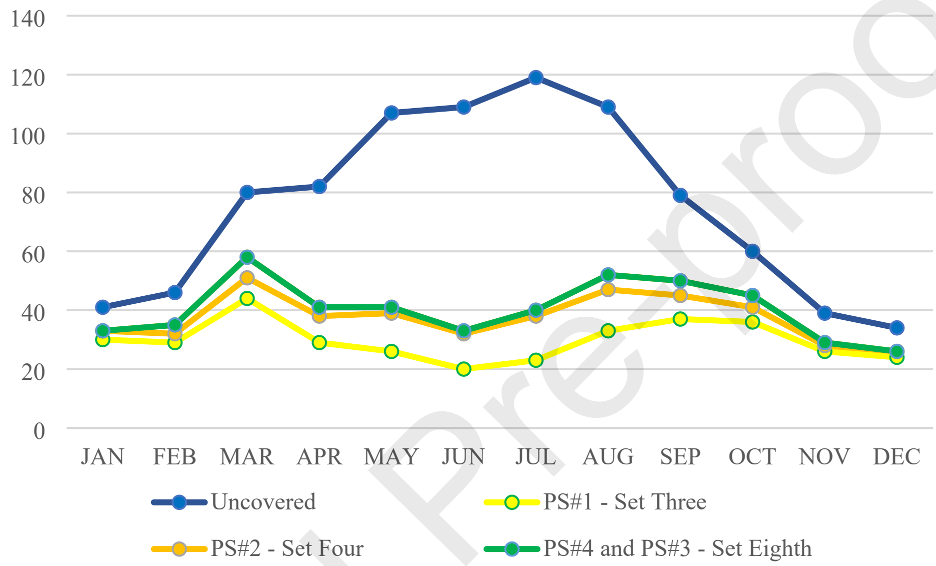
The lowest average air temperature in Évora, observed in January, is 10.1ºC (min. 5ºC to max. 14ºC), and the highest average occurs in August, at 26.5ºC (min. 16ºC to max. 34ºC), see Annex A, Figure A.1. However, under direct solar radiation on horizontal surfaces, especially during summer, these values could be higher, following 25.7/33.1kWh/m2/month in December/January and 72.6/68.9kWh/m2/month in July/August, due to its latitude of 38°34′00″ N, see Annex A, Figure A.2.

Relative humidity significantly impacts thermal comfort perception when falling outside of the 30 to 60% range. In Évora, the least humid month is July, with a low of 43.4%, and the most humid month is December, with a high of 76.4%. The figures from October to February fall within the UTCI comfort range, suggesting potential for user comfort, except for January, where the average is 10.1ºC, see Annex A, Figure A.1.

Évora shows low water vapor pressure values, as expressed by the LB Import EPW component, not exceeding the barometric pressure of 1080hPa. This minimal impact on thermal discomfort increases skin evaporation, facilitating body cooling through sweating and resulting in a cool sensation even under warm temperatures. Fluctuations around nine hPa are still comfortable, though slightly above the threshold of 6.8hPa [46], which can trigger migraines in some individuals, see Annex A, Figure A.4.

The LB Incident Radiation component generates incident radiation data, as shown in Figure 6, comparing the uncovered Patio figures to the eighth set, both crossed with PS#4. The highest value recorded in March was 58kWh/m2, lower than the uncovered Patio's 80kWh/m2 in the same month. As the TMS casts a shadow, excess energy is cut, as depicted by the gap between the blue (uncovered) and green line (eighth set), reducing UTCI figures. The TMS PS#1 set three and PS#2 set four show lower incident radiation than PS#3 and PS#4 set eight. PS#4 set eight was selected for its best performance due to its pavement thermal mass capacity, which can collect higher thermal energy when less comfortable. The values eventually converge between October and February, with ΔIRs of 15kWh/m2 (Oct), 10kWh/m2 (Nov), 8kWh/m2 (Dec & Jan), and 11kWh/m2 (Feb). The uncovered Patio contributes to higher UTCI figures, while the eighth set smooths the thermal sensation for users, evidenced by the significant ΔIR of 422kWh/m2.

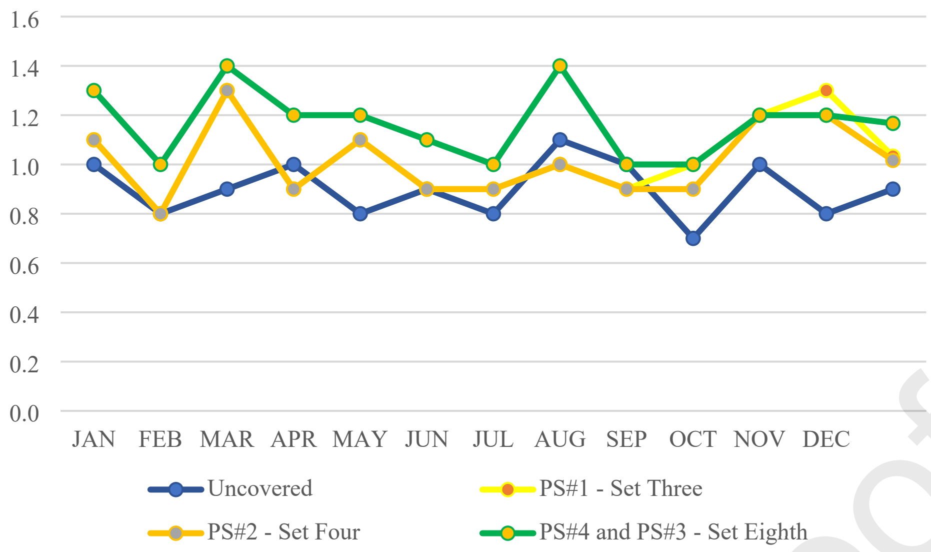
Figure 6 — Incident Radiation (kWh/m²) per month.



The LB Butterfly component generates the wind speed data shown in Figure 7. February, July, September, and October have the lowest average wind speed of 1.0 m/s, while March and August have the highest average wind speed of 1.4 m/s, see Annex A, Figure A.5. This helps balance the highest thermal sensation following the Portuguese climate pattern.

The uncovered Patio exhibits a lower average wind speed of 0.7 m/s in October and a higher average of 1.1 m/s in August (nortada), falling below the eighth set figure and reducing the warm thermal feeling. This situation results from the tunnel effect created by the wind speed, as indicated by the green line (eighth set) alongside the blue line (uncovered), which only aligns in September, as shown in Figure 7. Generally, the average wind speeds are similar in both scenarios, with a deviation of 0.3 m/s. The higher wind speed observed in the eighth set (PS#3 and PS#4) compared to the third (PS#1) and fourth (PS#2) is due to the higher enclosure, enhancing the efficiency of the tunnel effect, as depicted in Figure 7.

Figure 7 — Wind Speed (m/s) per month.



5. Discussion

Lightweight solutions like TMS prioritize free-form double curve design over UTCI readings to demonstrate their thermal efficiency and shadow performance. This approach supports the study's aim to enhance the thermal value of public outdoor spaces and improve user experience, consistent with findings from other research [9], [11], [28], [35], [47].

The study's methodology aligns with previous research, such as that by Elnokaly, Ayoub, and Ragheb [28][29], using computational techniques and energy analysis software to assess thermal behavior in challenging climates. However, this study stands out by focusing on a unique set of overlapping laminae and incorporating recent local climate data, thereby increasing the complexity and relevance of the research.

A comprehensive analysis of fourteen TMS configurations over a year provides insights into the optimal configuration for achieving the best outdoor thermal comfort (OTC). The results are consistent with recent studies, like Abbas and Khalid's [32], emphasizing the benefits of shading devices, high albedo surfaces, vegetation, and water features in improving urban OTC. This study, however, specifically focuses on TMS's impact on OTC.

5.1. UTCI-based methodology validation and performance

The study employed a systematic methodology, conducting 720 simulations to refine configurations based on their performance. The evaluation of covered and uncovered Patio solutions across four surface configurations confirmed the superior performance of TMS sets. To deepen this understanding, the team conducted an additional 84 specific simulations, a methodology echoed by others [32], [49], [50].

The analysis of UTCI and its components identified the best-performing TMS sets, optimizing summer comfort without compromising winter solar performance. The findings validate the use of software-based simulations and provide insights into outdoor solution performance [51]. Despite its success, the study highlights the simplified design of the proposed TMS, consisting of shadow bands on a double curvature shell (shown in Figure 3), reflecting the simplest design available from manufacturers [13], [10].

The study also illustrates the effectiveness of combining high thermal mass delay in pavements with TMS's shadow performance. This approach benefits indoor comfort and energy efficiency in maritime containers, reducing radiation exposure to passive elements and extending the lifespan of active systems [47], [52].

All simulations aimed to maximize Patio comfort during daylight hours (9 am to 5 pm), fostering public use of outdoor spaces. This approach aligns with findings that promote outdoor activities and social interactions [53], [54].

5.2. Urban outdoor comfort under UTCI assessment

The study underscores the challenge of achieving optimal outdoor comfort in partially wind-protected areas like Patios covered by TMS in southern Europe. The results demonstrate an improvement in outdoor comfort levels based on UTCI readings within the comfort range of >9ºC to <26ºC, aligning with related studies [9], [16], [55]. Additionally, the study suggests using EnergyPlus-based software or similar tools to support the analysis of TMS's UTCI ΔT, considering factors like angles, lengths, repetitions, and pavement surface specifications, also explored by other research [12], [50].

From a UTCI perspective, the eighth set, including PS#4, shows a UTCI deviation of ΔT = 3.4ºC (<9ºC) during winter and a correction of ΔT = 0.1ºC (>26ºC) during summer, resulting in a total ΔT = 3.5ºC (>9º to <26ºC) compared to the uncovered set's ΔT = 11.6ºC (>9º to <26ºC). This represents an average reduction of ΔT = 8.1ºC (>9º to <26ºC) annually, validating the heat stress reduction under the best-performing Patio variation [14], [16].

5.3. TMS design and reliability

However, the study acknowledges challenges posed by the membrane's steeper slopes in handling dynamic loads, such as north winds, which peak at 3.3 m/s during summer and 5.5 m/s in winter. The layout and container height help mitigate these issues. Even during sea storms, Évora, an inland low-altitude region, experiences maximum wind speeds of 18.05 m/s, well below the TMS limit of 36 m/s [57].

From a wind speed perspective, the eighth Patio set conforms to the Lawson classification, with wind speeds ranging from 1.2 to 1.4 m/s at 1 meter above the surface, as shown in Table 12. This level facilitates low metabolic activities like prolonged sitting [58], representing a reduction from regional speeds, even with the U-patio shape's two "open" corners. These findings align with other studies, particularly during warmer summer hours [59], [60].

Table 12 — Wind speed classification based on pedestrian comfort [58].

Threshold of wind speed	Quality class	Original description	Reference activity
U > 1.8 m/s	A	Covered area	Sitting long
U > 3.6 m/s	B	Pedestrians standing around	Sitting short

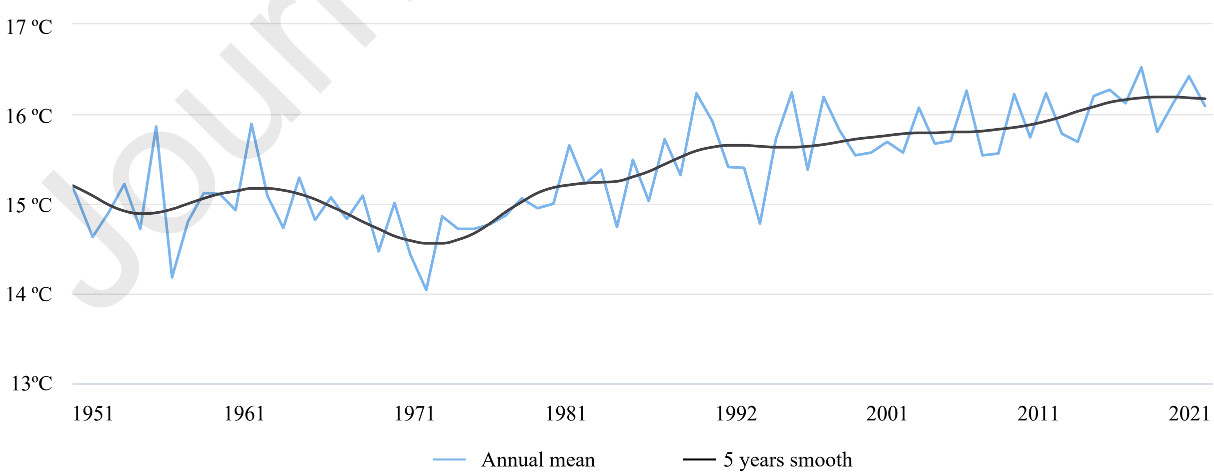
U > 5.3 m/s	C	Pedestrian walkthrough	Strolling
U > 7.6 m/s	D	Roads and car parks	Walking fast
U > 15.0 m/s	E	Dangerous	Unacceptable

Although there is a tie between sets eight and nine, set eight is smaller, uses less material, involves fewer repetitions, and impacts dynamic loads less. Therefore, from economic, construction, maintenance, and durability perspectives, it represents the more pertinent option.

5.4. Contribution to knowledge

To effectively address the impact of rising ambient temperatures and the formation of heat islands, understanding the significance of outdoor shadowing assessment is crucial in guiding the planning and design of public gathering areas. Like our own, several studies offer insights into optimizing shading solutions provided by TMS and mitigating the adverse effects of extreme heat on outdoor spaces [16], [17], [34], [35]. Utilizing outdoor spaces can provide users with more sustainable and comfortable environments by offering high-performance shading strategies, enhancing social interactions, and fostering community well-being [54]. While limited to Évora, Portugal, the principles and solutions behind this study are replicable elsewhere, especially in dense urban areas affected by extreme solar radiation. An example of this is the rising temperatures in continental Portugal as depicted in Figure 8. The black line indicates the 5-year smooth temperatures in Portugal, while the blue line shows the annual fluctuations in mean temperature. The peaks on the blue line represent the heatwaves recorded in Portugal's continental territory between 1951 and 2021, in line with global readings.

Figure 8 — The annual mean temperature observed in Portugal continental from 1951-2021[61].



5.5. Study limitations

The conditions and parameters were assessed virtually without fieldwork or a data collection campaign. Based on monthly UTCI average figures, the detailed information obtained from the study may be limited compared to methodologies that consider hourly or daily data.

The research highlighted in Ibrahim et al. (2020) comparing Ladybug Tools and ENVI-met shows that both tools maintain acceptable accuracy during typical daylight hours (8 am to 5 pm). However, Ladybug Tools is notably more efficient in processing times, allowing for rapid assessment of multiple design variations, a critical advantage in optimization studies. In this study, the authors stated that Ladybug Tools' results have shown a substantial level of agreement with ENVI-met ($R^2 = 0.97$) in three different urban configurations [62].

The study methods can be reapplied to locations with similar climate conditions, particularly those inland, at low altitudes, and with prevalent north-south wind patterns along the Mediterranean coast in cities such as Seville, Murcia, and Zaragoza (Spain); Nimes (France); Foggia (Italy); and Athens (Greece). However, applying this model elsewhere requires adjustments. The study has demonstrated that it reliably identifies the key geometry characteristics for TMS, including the number of lanes, overlays, and angles, according to local conditions.

Retractable solutions were initially considered but dismissed due to higher costs associated with designing, construction/installation, operation, maintenance, and durability. While popular in South Mediterranean countries, they tend to cover mostly narrow streets, limiting their scope of application. Instead, our study focused on an affordable and reliable shading solution that does not require constant adjustments to solar conditions and wind speeds/gusts in open fields.

6. Conclusion: submission of contributions

The research addresses the complexity of the urban microclimate and outdoor thermal comfort (OTC) in inland, low-altitude Mediterranean cities. It focuses on optimizing TMS shading across four pavement types to enhance OTC in urban environments, supported by a robust methodological framework that includes computational fluid dynamics (CFD) and EnergyPlus simulations using Universal Thermal Climate Index (UTCI) readings for assessing user thermal comfort. The findings demonstrate that the geometry, shape, orientation, and passive cooling measures of TMS are crucial in shaping microclimate conditions that enhance users' thermal comfort, primarily by promoting evaporative cooling and increasing air velocity around the human body.

The research explores various TMS designs on a U-shaped Patio enclosed by maritime containers. Fourteen configurations with different slopes, overlap angles, lengths, and repetitions, along with four thermal mass pavement compositions, were tested:

The study identified the eighth TMS set as the best-performing, covering the Patio with an entire concrete surface (PS#4). This set features a slope angle of 38.2º, an overlap angle of 74.4º on the summer solstice, a length of 3.88m, and five tensile membrane lanes. It aimed to minimize the UTCI ΔT between the covered and uncovered Patio, reducing winter solar radiation by ΔT = 1.1ºC (<9ºC) and significantly lowering high summer temperatures by ΔT = 9.4ºC (>26ºC). On a monthly average, the eighth set achieved a decrease of ΔT = 8.3ºC (>9ºC to <26ºC) compared to the uncovered Patio.

The research found that TMS sets with slopes around 40º and overlap angles around 75º, considering the latitude of 38º, produce fewer repetitions upon extending each lane by 12.8% in length, positively impacting thermal user comfort.

The results emphasize the importance of finding optimal shadowing structures to enhance user comfort in regions with warm, temperate, and dry, hot summers. The study also showcases the potential of software to find effective and high-performance outdoor solutions based on UTCI readings, thus positively impacting public socialization.

When combined with various pavement compositions, the use of white-colored polytetrafluoroethylene (PTFE) TMS contributes to improved Patio performance by reflecting solar radiation and providing non-combustibility, beneficial for both user comfort and safety.

The straightforward methodology enables TMS thermal evaluation studies using dynamic simulation software, EnergyPlus 2022, part of LB 1.5 for environmental analysis. The simulation parameters, including the geometry of the TMS and their configurations, were derived from different TMS configurations, as shown in Tables 1 and 2. The materials used, such as white-colored PTFE fiberglass fabric coated with Teflon®, were chosen for their durability and reflectivity, adhering to ASTM E424 standards. The simulation environment was modeled in Rhinoceros 7, reflecting the typical housing structure of southern Europe. To enhance the reliability of our findings, we have provided comprehensive descriptions of the membrane angles, lengths, and repetitions, as well as the pavement sets with their respective characteristics and properties. This is intended to ease the reproduction of our study under similar conditions, allowing for the validation of our results by other researchers in the field.

Furthermore, the study highlights the need for establishing outdoor thermal performance thresholds in the codes for tensile membrane and other shadowing structures, such as "ASCE/SEI 55-16," "Eurocode 1: Actions on structures," or similar regulations, which should be addressed in forthcoming updates. As a next step, the team will explore Python programming to automate the form-finding analysis process using the same software, defining the slope, inclination angles, length, and repetitions while incorporating various pavement surfaces without modeling and simulating each set repeatedly until achieving the best performance.

The future research trajectory for optimizing OTC should encompass expanding geographical studies to validate effectiveness in diverse weather conditions, evaluating urban solutions' long-term durability and maintenance needs for a comprehensive assessment of their viability and cost-effectiveness over time. Additionally, investigating the use of materials focusing on enhanced properties, features, and composition, and assessing the impact of adding vegetation, water elements, and albedo ratios to improve OTC.

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Declaration of generative AI and AI-assisted technologies in the writing process

While preparing this work, the authors used ChatGPT v3.5/OpenAI to improve language and readability. After using this tool/service, the authors reviewed and edited the content as needed and took full responsibility for the publication's content.

References

[1] E. Piguet, "Environment and Migration," in *International Encyclopedia of Human Geography (Second Edition)*, Second Edition., A. Kobayashi, Ed., Oxford: Elsevier, 2020, pp. 163–168. doi: https://doi.org/10.1016/B978-0-08-102295-5.10258-6.

[2] C. Santos-Rojo, M. Llopis-Amorós, and J. M. García-García, "Overtourism and sustainability: A bibliometric study (2018–2021)," *Technol. Forecast. Soc. Change*, vol. 188, p. 122285, 2023, doi: https://doi.org/10.1016/j.techfore.2022.122285.

[3] S. Dotsey and A. Lumley-Sapanski, "Temporality, refugees, and housing: The effects of temporary assistance on refugee housing outcomes in Italy," *Cities*, vol. 111, p. 103100, 2021, doi: https://doi.org/10.1016/j.cities.2020.103100.

[4] Köppen, W., "The Geographical System of Climates," *Geogr. J.*, no. 88(5), pp. 414–417, 1936.

[5] Kottek, M., Grieser, J., Rubel, F., Beck, C., and Rudolf, B., "World Map of the Köppen-Geiger climate classification updated," *Meteorol. Z.*, vol. 15(3), pp. 259–263, 2006.

[6] Instituto Nacional de Estatística - Censos 2021, *XVI Recenseamento Geral da População*, vol. VI Recenseamento Geral da Habitação: Resultados definitivos. INE, 2022, 2022. [Online]. Available: https://www.ine.pt/xurl/pub/65586079

[7] Birdair, "Tensile Membrane Structures: Energy Use and Lighting." BIRDAIR, 2015. [Online]. Available: https://www.birdair.com/files/brochures/Tensile%20Membrane%20Structures%20-%20Energy%20Use%20and%20Lighting.pdf

[8] A. M. Elnokaly, J. Chilton, and R. Wilson, "Environmental Behaviour of Tensile Membrane Structures," in *Energy, Water, and Environment*, 2002. [Online]. Available: https://citeseerx.ist.psu.edu/document?doi=8e8918186e88cd41e5c5eebb5c3b2a6e47c43df3&repid=rep1&type=pdf

[9] M. Pankow, B. Justusson, M. Riosbaas, A. M. Waas, and C. F. Yen, "Effect of fiber architecture on tensile fracture of 3D woven textile composites," *Compos. Struct.*, vol. 225, p. 111139, 2019, doi: https://doi.org/10.1016/j.compstruct.2019.111139.

[10] A. L. Marbaniang, S. Dutta, and S. Ghosh, "Tensile Membrane Structures: An Overview," in *Advances in Structural Engineering*, K. V. L. Subramaniam and M. A. Khan, Eds., Singapore: Springer Singapore, 2020, pp. 29–40.

[11] D. Collins and S. L. Stadler, "Public Spaces, Urban," in *International Encyclopedia of Human Geography (Second Edition)*, Second Edition., A. Kobayashi, Ed., Oxford: Elsevier, 2020, pp. 103–111. doi: https://doi.org/10.1016/B978-0-08-102295-5.10212-4.

[12] Y. Chen, M. Li, J. Cao, S. Cheng, and R. Zhang, "Effect of climate zone change on energy consumption of office and residential buildings in China," *Theor. Appl. Climatol.*, Feb. 2021, doi: 10.1007/s00704-021-03544-w.

[13] American Society of Civil Engineers, *ASCE Subject Headings: Tensile structures, Standards and codes, Membranes, Structural analysis, Cables, Fabric structures,*

Construction materials, Professional societies. © American Society of Civil Engineers, 2016. [Online]. Available: https://doi.org/10.1061/9780784414378

[14] European Committee for Standardization (CEN), *Eurocode 1: Actions on structures – Part 1-4: General actions – Wind actions*. Brussels, Belgium: European Committee for Standardization, 2019.

[15] P. Bröde *et al.*, "Deriving the operational procedure for the Universal Thermal Climate Index (UTCI)," *Int. J. Biometeorol.*, vol. 56, no. 3, pp. 481–494, May 2012, doi: 10.1007/s00484-011-0454-1.

[16] E. Tapias and S. Soni, *Building-up urban open spaces from shadow range analyses*. 2014. doi: 10.52842/conf.ecaade.2014.1.129.

[17] ASHRAE, "ANSI/ASHRAE STANDARD 55 – Thermal environmental conditions for human occupancy." American Society of Heating, Refrigerating and Air-Conditioning Engineers, 2020. [Online]. Available: https://www.ashrae.org/technical-resources/bookstore/standard-55-thermal-environmental-conditions-for-human-occupancy

[18] R. Aghamolaei, M. M. Azizi, B. Aminzadeh, and J. O'Donnell, "A comprehensive review of outdoor thermal comfort in urban areas: Effective parameters and approaches," *Energy Environ.*, vol. 34, no. 6, pp. 2204–2227, Sep. 2023, doi: 10.1177/0958305X221116176.

[19] J. Li, J. Niu, C. M. Mak, T. Huang, and Y. Xie, "Exploration of applicability of UTCI and thermally comfortable sun and wind conditions outdoors in a subtropical city of Hong Kong," *Sustain. Cities Soc.*, vol. 52, p. 101793, Jan. 2020, doi: 10.1016/j.scs.2019.101793.

[20] A. Santos Nouri, J. P. Costa, M. Santamouris, and A. Matzarakis, "Approaches to Outdoor Thermal Comfort Thresholds through Public Space Design: A Review," *Atmosphere*, vol. 9, no. 3, Art. no. 3, Mar. 2018, doi: 10.3390/atmos9030108.

[21] Y. Dzyuban *et al.*, "Outdoor thermal comfort research in transient conditions: A narrative literature review," *Landsc. Urban Plan.*, vol. 226, p. 104496, Oct. 2022, doi: 10.1016/j.landurbplan.2022.104496.

[22] M. H. Elnabawi and N. Hamza, "Behavioural Perspectives of Outdoor Thermal Comfort in Urban Areas: A Critical Review," *Atmosphere*, vol. 11, no. 1, Art. no. 1, Jan. 2020, doi: 10.3390/atmos11010051.

[23] F. Sarhadi and V. B. Rad, "The structural model for thermal comfort based on perceptions individuals in open urban spaces," *Build. Environ.*, vol. 185, p. 107260, Nov. 2020, doi: 10.1016/j.buildenv.2020.107260.

[24] E. Antonini, V. Vodola, J. Gaspari, and M. De Giglio, "Outdoor Well-being and Quality of Life: A Scientific Literature Review on Thermal Comfort," *Energies*, vol. 13, no. 8, 2020, doi: 10.3390/en13082079.

[25] G. Pérez, J. Coma, and L. F. Cabeza, "Chapter 3.1 - Vertical Greening Systems to Enhance the Thermal Performance of Buildings and Outdoor Comfort," in *Nature Based Strategies for Urban and Building Sustainability*, G. Pérez and K. Perini, Eds., Butterworth-

Heinemann, 2018, pp. 99–108. doi: https://doi.org/10.1016/B978-0-12-812150-4.00009-4.

[26] E. Eugenio Samano Baca and I. Tsung Tsai, "A Model-Based Approach to Measuring the Effect of Shading on Outdoor Thermal Comfort," *Int. J. Eng. Technol.*, vol. 7, no. 2, pp. 116–121, Apr. 2015, doi: 10.7763/IJET.2015.V7.777.

[27] M. H. Elnabawi and N. Hamza, "Outdoor Thermal Comfort: Coupling Microclimatic Parameters with Subjective Thermal Assessment to Design Urban Performative Spaces," *Buildings*, vol. 10, no. 12, 2020, doi: 10.3390/buildings10120238.

[28] A. M. Elnokaly, "CFD Analysis of Tensile Conical Membrane Structures as Microclimate Modifiers in Hot Arid Regions," *Civ. Eng. Archit.*, vol. 2, no. 2, Art. no. 2, Feb. 2014, doi: 10.13189/cea.2014.020204.

[29] M. Ayoub and S. A. Ragheb, "Parametric investigation of the solar-thermal performance of tensile membrane roofs in hot-desert climates," *Energy Build.*, vol. 297, p. 113449, Oct. 2023, doi: 10.1016/j.enbuild.2023.113449.

[30] A. S. Nouri, A. Lopes, J. P. Costa, and A. Matzarakis, "Confronting potential future augmentations of the physiologically equivalent temperature through public space design: The case of Rossio, Lisbon," *Sustain. Cities Soc.*, vol. 37, pp. 7–25, Feb. 2018, doi: 10.1016/j.scs.2017.10.031.

[31] A. Matzarakis, H. Mayer, and M. G. Iziomon, "Applications of a universal thermal index: physiological equivalent temperature," *Int. J. Biometeorol.*, vol. 43, no. 2, pp. 76–84, Oct. 1999, doi: 10.1007/s004840050119.

[32] M. V. Rodríguez, S. G. Melgar, and J. M. A. Márquez, "Assessment of aerial thermography as a method of in situ measurement of radiant heat transfer in urban public spaces," *Sustain. Cities Soc.*, vol. 87, p. 104228, 2022, doi: https://doi.org/10.1016/j.scs.2022.104228.

[33] Z. R. Abaas and Z. Khalid, "Towards local sustainability: A case study to evaluate outdoor urban spaces in Baghdad using physiological equivalent temperature index," *City Environ. Interact.*, vol. 20, p. 100115, Dec. 2023, doi: 10.1016/j.cacint.2023.100115.

[34] S. Jafarian, E. Sarkardehee, D. Monsefi Parapari, and M. R. Mojahedi, "Introducing an Optimal Model and Dimensions of Lightweight Membrane Canopy for Hot and Dry Climate of City of Semnan in Iran," *Renew. Energy Res. Appl.*, vol. 2, no. 1, pp. 101–110, Jan. 2021, doi: 10.22044/rera.2021.10512.1048.

[35] I. Elwy, Y. Ibrahim, M. Fahmy, and M. Mahdy, "Outdoor microclimatic validation for hybrid simulation workflow in hot arid climates against ENVI-met and field measurements," *5th Int. Conf. Energy Environ. Res. ICEER 2018 23-27 July 2018 Prague Czech Repub.*, vol. 153, pp. 29–34, Oct. 2018, doi: 10.1016/j.egypro.2018.10.009.

[36] L. Sultana and N. Bari, "A Study on the Impact of Tensile Fabric Structure in Semi-Outdoor Spaces of Tropical Cities," *Int. J. New Innov. Eng. Techno*, vol. 10, no. 4, 2019.

[37] L. Chen, Y. Zhang, J. Han, and X. Li, "An investigation of the influence of ground surface properties and shading on outdoor thermal comfort in a high-altitude residential area,"

Front. Archit. Res., vol. 10, no. 2, pp. 432–446, Jun. 2021, doi: 10.1016/j.foar.2020.12.005.

[38] "ladybug_comfort.utci module — ladybug comfort documentation." Accessed: July 6th, 2023. [Online]. Available: https://www.ladybug.tools/ladybug-comfort/docs/ladybug_comfort.utci.html

[39] COST Action 730, "The Universal Thermal Climate Index UTCI for Assessing the Thermal Environment of the Human Being," International Society of Biometeorology, Paris, 2009. Accessed: July 24th, 2023. [Online]. Available: (https://www.ladybug.tools/ladybug-comfort/docs/ladybug_comfort.utci.html). The clothing behavior of individuals, specifically the amount of insulation they choose to wear, is influenced by the air temperature in a non-linear manner within the range of 0.4 clo to 3.0 clo (http://utci.org/public/UTCI%20Executive%20Summary.doc). To calculate the UTCI, four variables are required: air temperature, dew point temperature (or relative humidity), both at 2m above ground level, wind speed at 10m above ground level, and mean radiant temperature (MRT), see Table 2 (https://climate-adapt.eea.europa.eu/en/metadata/indicators/thermal-comfort-indices-universal-thermal-climate-index-1979-2019).

[40] European Climate Adaptation Platform, "Thermal Comfort Indices - Universal Thermal Climate Index, 1979-2020 — English." Accessed: July 6th, 2023. [Online]. Available: https://climate-adapt.eea.europa.eu/en/metadata/indicators/thermal-comfort-indices-universal-thermal-climate-index-1979-2019

[41] P. Cohen, O. Potchter, and A. Matzarakis, "Human thermal perception of Coastal Mediterranean outdoor urban environments," *Appl. Geogr.*, vol. 37, pp. 1–10, 2013, doi: https://doi.org/10.1016/j.apgeog.2012.11.001.

[42] Psikuta, A. *et al.*, validation of the Fiala multi-node thermophysiological model for UTCI application. International journal of biometeorology, 56(3), 443–460. https://doi.org/10.1007/s00484-011-0450-5," *Int. J. Biometeorol.*, vol. 56(3), pp. 443–460, 2012, doi: https://doi.org/10.1007/s00484-011-0450-5.

[43] P. Brode *et al.*, "Calculating UTCI equivalent temperatures," in *Poster presentation*, Boston, Massachusetts, USA: ICEE 2009, 7/8 2009. Accessed: March 23rd, 2023. [Online]. Available: http://www.utci.org/utci_poster.pdf

[44] UNESCO, "Historic Centre of Évora," World Heritage Convention. [Online]. Available: https://whc.unesco.org/en/list/361

[45] "The Engineering ToolBox." Accessed: July 24th, 2023. [Online]. Available: https://www.engineeringtoolbox.com/

[46] Anders Hougaard, Faisal Amin, Anne Werner Hauge, Messoud Ashina, and Jes Olesen, "Provocation of migraine with aura using natural trigger factors," *Neurology*, vol. 80, no. 5, p. 428, Jan. 2013, doi: 10.1212/WNL.0b013e31827f0f10.

[47] S. Lu, H. Qian, Q. Guo, and M. Gu, "Energy-saving performance and mechanism studies on membrane ceiling: A case study of the National Speed Skating Oval," *Energy Sustain. Dev.*, vol. 72, pp. 88–99, 2023, doi: https://doi.org/10.1016/j.esd.2022.11.004.

[48] Abbas Ibrahim Mu'azu, "Sustainable design strategy: assessment of the impact of design variables on energy consumption of office buildings in Abuja, Nigeria."

[49] A. S. Nouri, A. Lopes, J. P. Costa, and A. Matzarakis, "Confronting potential future augmentations of the physiologically equivalent temperature through public space design: The case of Rossio, Lisbon," *Sustain. Cities Soc.*, vol. 37, pp. 7–25, 2018, doi: https://doi.org/10.1016/j.scs.2017.10.031.

[50] A. Castell and C. Solé, "11 - Design of latent heat storage systems using phase change materials (PCMs)," in *Advances in Thermal Energy Storage Systems*, L. F. Cabeza, Ed., in Woodhead Publishing Series in Energy. , Woodhead Publishing, 2015, pp. 285–305. doi: https://doi.org/10.1533/9781782420965.2.285.

[51] Y. Ibrahim, T. Kershaw, and P. Shepherd, *A methodology For Modelling Microclimates: A Ladybug-tools and ENVI-met verification study*. 2020.

[52] N. D. Cortiços, "Improving residential building efficiency with membranes over façades: The Mediterranean context," *J. Build. Eng.*, vol. 32, p. 101421, Nov. 2020, doi: 10.1016/j.jobe.2020.101421.

[53] W. Liu, Y. Zhang, and Q. Deng, "The effects of urban microclimate on outdoor thermal sensation and neutral temperature in hot-summer and cold-winter climate," *Energy Build.*, vol. 128, pp. 190–197, 2016, doi: https://doi.org/10.1016/j.enbuild.2016.06.086.

[54] S. Jombach, L. Huawei, G. Wang, I. Valánszki, and K. Filepné Kovács, *Greenway Exploration in the Satellite Jungle: Discovery of Urban and Rural Green Network with Satellite Image Analysis in Hungary*. 2019. doi: 10.7275/wvhx-4078.

[55] P. Bröde, E. Krüger, and D. Fiala, "UTCI: Validation and practical application to the assessment of urban outdoor thermal comfort," *Geogr. Pol.*, vol. 86, pp. 11–20, Jan. 2013, doi: 10.7163/GPol.2013.2.

[56] H. Lee, H. Mayer, and W. Kuttler, "Impact of the spacing between tree crowns on the mitigation of daytime heat stress for pedestrians inside E-W urban street canyons under Central European conditions," *Urban For. Urban Green.*, vol. 48, p. 126558, 2020, doi: https://doi.org/10.1016/j.ufug.2019.126558.

[57] "Climate in Évora, Portugal," Climate-Data.org. [Online]. Available: https://en.climate-data.org/europe/portugal/evora/evora-548/.

[58] T. V. Lawson, "The wind content of the built environment," *J. Wind Eng. Ind. Aerodyn.*, vol. 3, no. 2, pp. 93–105, 1978, doi: https://doi.org/10.1016/0167-6105(78)90002-8.

[59] H. Jaafar, I. Lakkis, and A. Yeretzian, "Impact of boundary conditions in a microclimate model on mitigation strategies affecting temperature, relative humidity, and wind speed in a Mediterranean city," *Build. Environ.*, vol. 210, p. 108712, 2022, doi: https://doi.org/10.1016/j.buildenv.2021.108712.

[60] J. A. Acero and J. Arrizabalaga, "Evaluating the performance of ENVI-met model in diurnal cycles for different meteorological conditions," *Theor. Appl. Climatol.*, vol. 131, no. 1–2, pp. 455–469, Jan. 2018, doi: 10.1007/s00704-016-1971-y.

[61] Climate Change Knowledge Portal. Accessed: March 23rd, 2023. [Online]. Available: https://climateknowledgeportal.worldbank.org/country/portugal/climate-data-historical,

[62] Ibrahim, Yasser & Kershaw, Tristan & Shepherd, Paul. (2020). A methodology For Modelling Microclimates: A Ladybug-tools and ENVI-met verification study.

Optimizing outdoor comfort under a Universal Thermal Climate Index with membranes' shading systems: a case study report in Évora, Portugal

Highlights:

1. *Optimizing outdoor thermal comfort through the use of membranes' shading systems.*
2. *UTCI optimizes TMS design to improve users' outdoor comfort.*
3. *Balanced UTCI levels can promote outdoor inter-community socialization.*
4. *Current codes for outdoor shadowing structures do not address thermal performance.*
5. *Next BIM software should enable energy related form-finding automation.*

Annex A

The data presented in the figures below represent the average climatic conditions of Évora over the last decade, as gathered from the Andrew Marsh web tool[7]. The information has been extracted from the EPW file (2022), which contains hourly or sub-hourly data for the region.

Figure A.1 — Dry Bulb Temperature.

[7] https://drajmarsh.bitbucket.io/data-view2d.html, accessed on April 20, 2023.

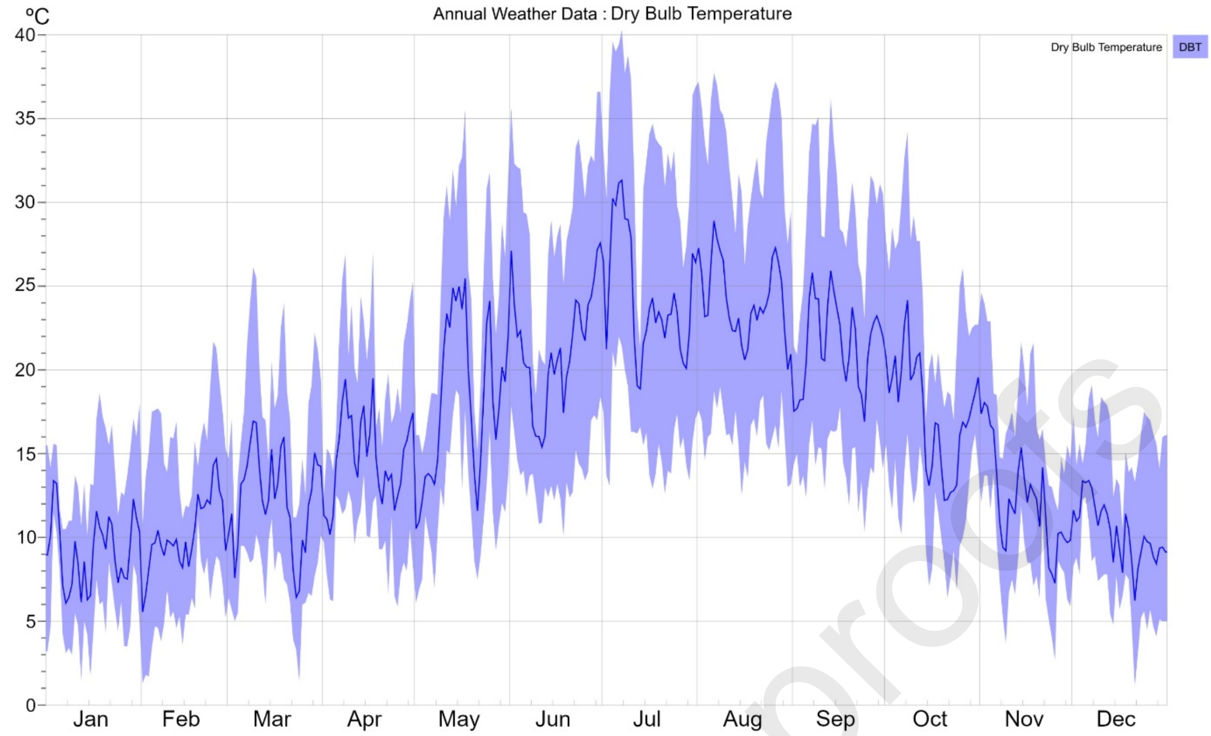


Figure A.2 — Global Horizontal Solar Radiation.

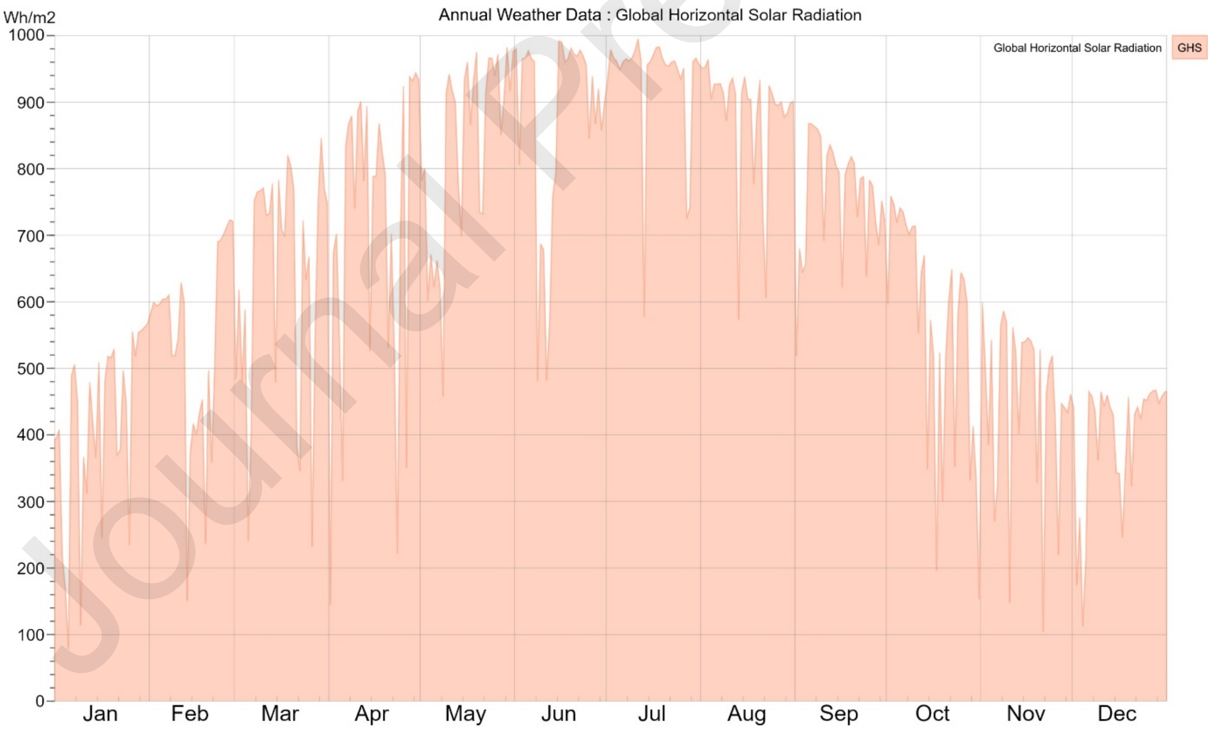


Figure A.3 — Relative humidity.

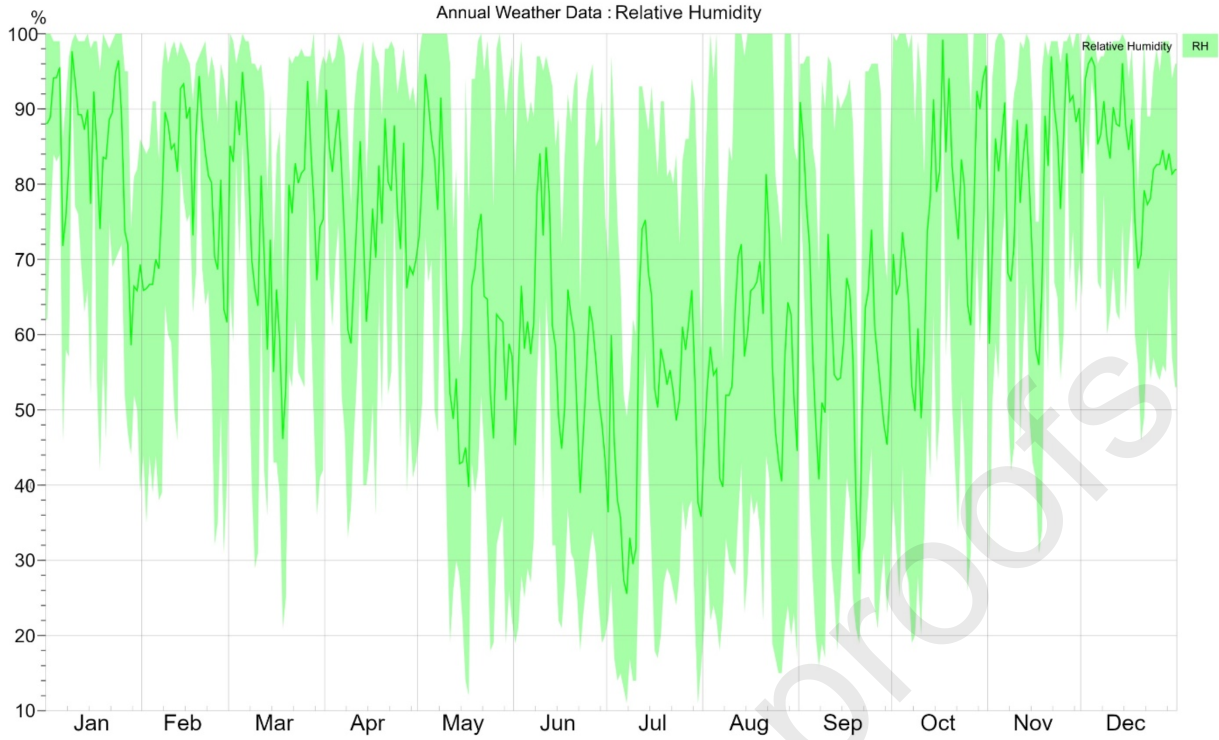


Figure A.4 — Atmospheric pressure.

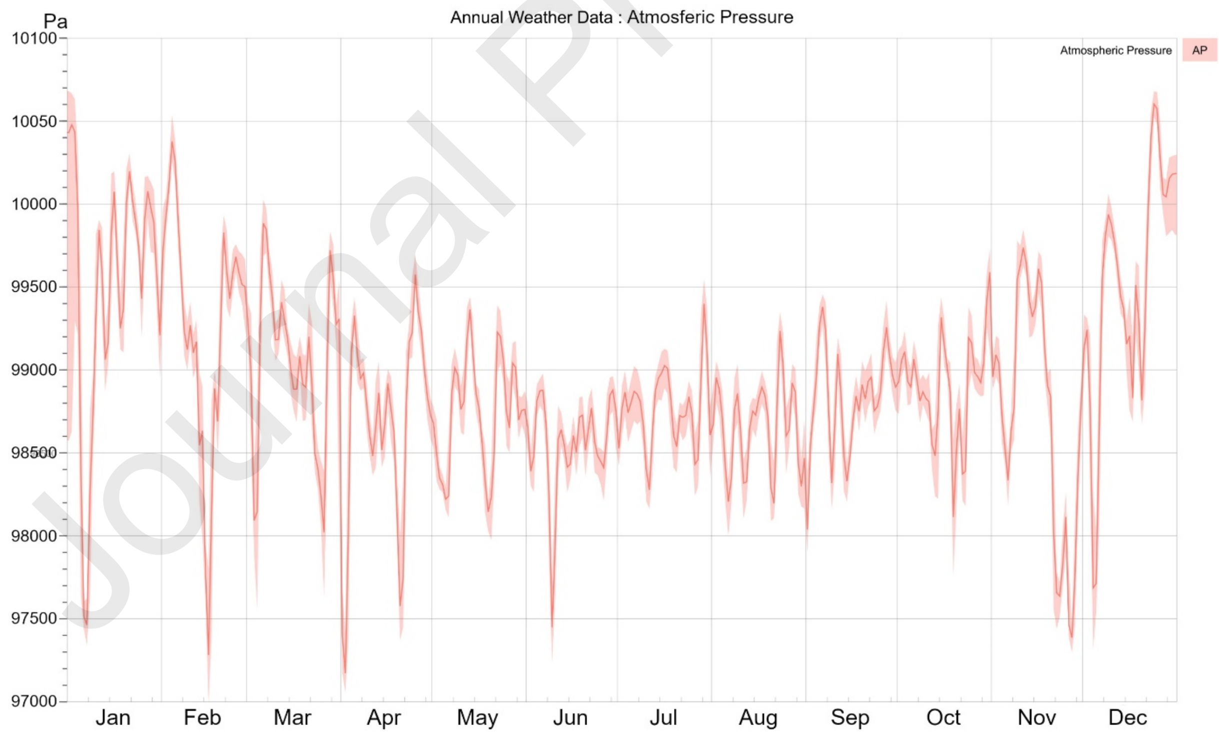


Figure A.5 — Wind Direction.

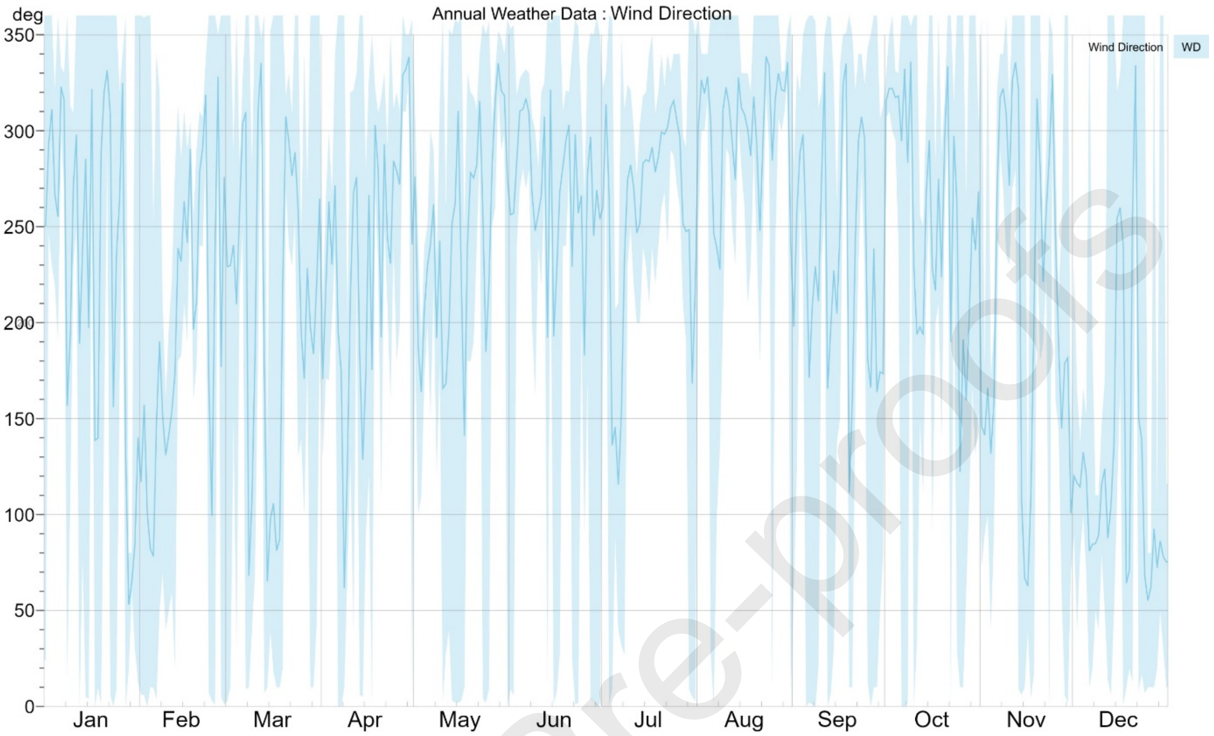
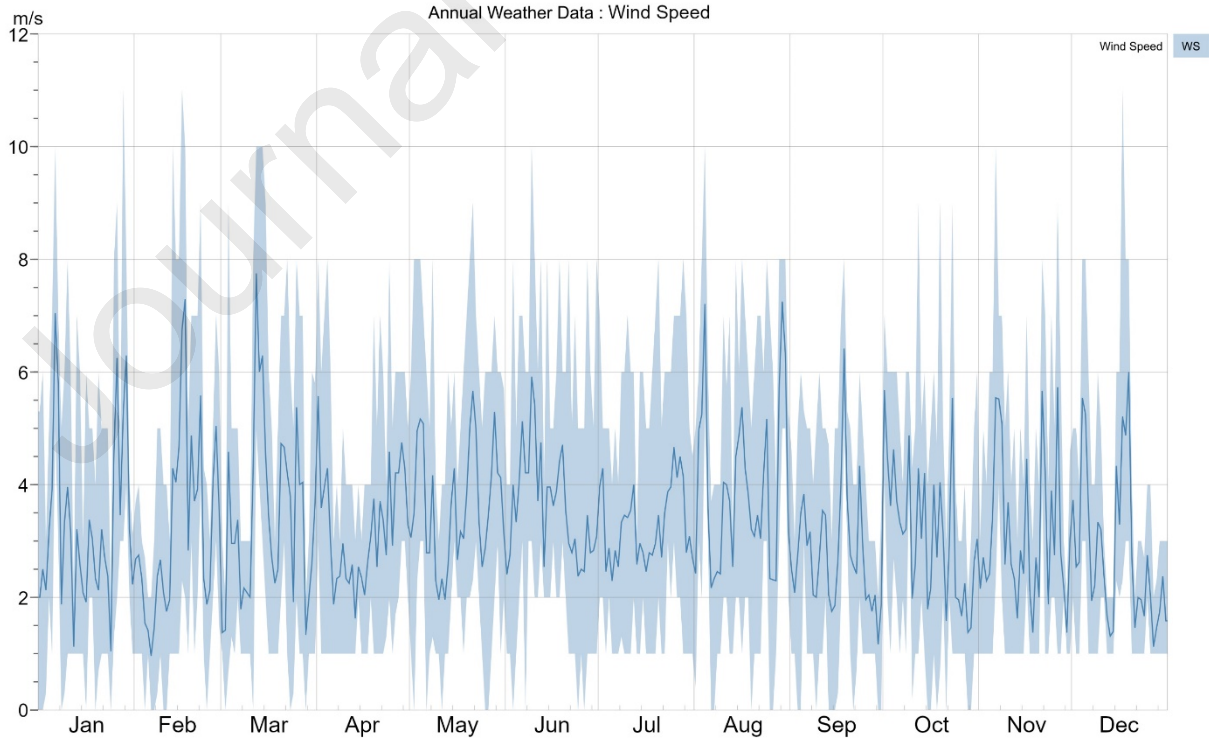


Figure A.6 — Wind Speed.



Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: